Early childhood interventions and their impact on poor children and their mothers: Evidence from Ecuador*

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Abstract

This paper evaluates two early childhood interventions in Ecuador, a home-visit program and a child care center program, funded through the same proposal-scoring mechanism. For each program we estimate effects on poor children and their mothers in a regression discontinuity design around the funding cutoff. The forcing variable is a proposal-level score that takes only 34 distinct values, so we assess every estimate with inference methods designed for discrete running variables and few clusters. Our most robust finding is that home visits raise children's overall development: the bias-aware honest confidence interval excludes zero, also when we double the smoothness bound, apply the fuzzy correction, or restrict the sample to an age range that is balanced across the cutoff. The point estimates further suggest that home visits reduce mothers' labor supply and stress, and that child care centers raise maternal working hours and income with mixed effects on children, but these estimates do not survive the most demanding inference procedures. The pattern of point estimates is consistent with a trade-off between child development and maternal labor supply; we present the cross-program comparison as descriptive because the two programs serve different populations. *JEL*-codes: J13, I28, H40, O12

*This version: August 2026. An earlier version of this paper circulated in 2011 as a Tinbergen Institute discussion paper but was never published. At the time we were reluctant to publish the results because we were uncertain whether the inference was valid: the forcing variable is a proposal-level funding score that takes only 34 distinct values, and it was not clear that the conventional standard errors we reported adequately reflected this discreteness and the small number of underlying clusters. This version resolves that concern. It re-estimates the design with modern local-polynomial regression-discontinuity methods and assesses the robustness of every conclusion using inference procedures developed for discrete running variables and few clusters (a wild cluster bootstrap, bias-aware honest confidence intervals, and finite-sample randomization inference), and reports explicitly which effects survive them. We acknowledge the financial support from the Ecuadorian government and the Inter-American Development Bank. The usual disclaimer applies. Rosero is the Director of the Statistics Division and Chief Statistician in FAO based in Rome-Italy. Oosterbeek is affiliated with the Vrije Universiteit Amsterdam and Tinbergen Institute.

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1 Introduction

This paper examines the impact of two early childhood interventions, home visits and child care centers, on cognitive, motor and health outcomes of young children from poor families in Ecuador, and on the parenting styles, psychological wellbeing and labor market outcomes of their mothers.

At the time of our study, the largest funding organization of early childhood programs in Ecuador ranked proposals of prospective providers of such programs on the basis of a score which was a mixture of perceived quality of the providers and the social background of the community served by the program. The organization then allocated its available budget to the proposals from high to low scores until its funds were depleted. This funding route has since been replaced, but both interventions remain in operation today (Section 2).

Our sample consists of children (and their mothers) who were listed in submitted proposals. Roughly half of the children in our sample were listed in proposals with a score that just fell short of the funding threshold. These form the control groups of the two interventions. The other half of the children in our sample were listed in proposals with a score that just exceeded the funding threshold. These form the treatment groups.

A large literature documents the effects of early childhood interventions. Reviews of the developed-country evidence are provided by Heckman (2006), Currie (2001) and Almond and Currie (2011). In developing countries, home-visit programs across six countries consistently find positive effects on child development (Engle et al., 2007; Walker et al., 2011; Grantham-McGregor et al., 2007), and these gains can extend well beyond the program years: the twenty-year follow-up of the Jamaican stimulation experiment finds large returns in adult earnings (Gertler et al., 2014), and a randomized trial in Colombia shows that home stimulation improves child development by raising parental investments (Attanasio et al., 2020). Programs taken to national scale tend to find more modest effects (Andrew et al., 2018; Carneiro et al., 2024). Center-based programs covering eleven developing countries similarly find positive child-development effects (Engle et al., 2007; Leroy et al., 2012; Berlinski et al., 2008), and center-based care reliably raises mothers' labor-market participation (Berlinski and Galiani, 2007; Berlinski et al., 2011; Halim et al., 2023). Broader syntheses are provided by Black et al. (2017) across the life course, by Jeong et al. (2021) for parenting interventions in low- and middle-income countries, and by Attanasio et al. (2022) and Attanasio (2026) from an economics perspective. The literature on early childhood programs in developing countries is thus large. Evidence that evaluates home visits and center-based care side by side, within the same institutional setting and with the same forcing variable and outcome measures, remains scarce; Sylvia et al. (2022) find that home-based delivery outperforms a center-based version of the same curriculum in rural China, but comparable evidence for large-scale national programs is limited.

Our study contributes in three ways to this literature. First, we consider a broad range of outcomes: children's cognitive and motor development, children's health, parenting styles, mothers' labor supply and income, and mothers' psychological wellbeing. Looking at this range of outcomes gives a more complete picture of the effects of early childhood interventions. Second, we evaluate two different early childhood interventions in parallel, using the same funding mechanism, the same sampling design, the same tests and questionnaires and the same estimation method. We stress that this is a parallel evaluation and not a direct comparison: the two programs were proposed for different communities and differ in intensity, delivery and the population served, so differences between the two sets of estimates cannot be attributed to program type alone. Third, in comparison to most other studies that evaluate home visit programs, we analyze a large scale national program instead of a small, tailor-made intervention and we collected data from a relatively large sample.

The main results are as follows. (i) Home visits raise children's overall development. This is the most robust finding: the bias-aware honest confidence interval excludes zero, and it continues to do so when we double the smoothness bound, when we apply the fuzzy correction of Noack and Rothe (2024), and when we restrict the sample to an age range that is balanced across the cutoff. The wild cluster bootstrap over the 34 proposals does not reject at conventional levels ($p = 0.36$), so some uncertainty remains also for this finding. (ii) Home visits also improve specific cognitive and fine-motor outcomes and mothers' psychological wellbeing while reducing mothers' labor-market participation. These effects are significant under conventional clustered inference but not under the wild cluster bootstrap or the honest confidence intervals, so we treat them as suggestive. (iii) For child care centers the estimated effects on children are mixed in sign and are not robust to the inference methods for the discrete running variable. (iv) The point estimates for child care centers show higher maternal labor supply and income, but, apart from the household head's earnings, these are not statistically significant even under conventional standard errors in the local-linear specification. The pattern of point estimates is consistent with a trade-off between child development and maternal labor supply, but the evidence for the individual components of this trade-off varies in strength, and the cross-program part of the pattern is descriptive.

We proceed as follows. The next section provides further details about the early childhood development programs in Ecuador that we study, and about the context. Section 3 discusses the empirical approach used to identify the programs' impacts. Section 4 describes the sampling design and the scales on which various outcomes are measured. Section 5 provides descriptive statistics of the main variables used in the analysis. It also present evidence supporting the assumption that the groups above and below the funding threshold are not systematically different. Section 6 presents and discusses the main results. Section 7 summarizes and concludes.

2 Context and interventions

2.1 Context

Ecuador is an upper-middle income country, characterized by high poverty and high inequality. Of its total population of about 18 million people (2025), some 7.6 percent are children under five years old (2022 census), on the order of 1.4 million. National income poverty stood at 25.5 percent in 2024, and it is considerably higher among young children: more than half of children under five are poor on the unsatisfied-basic-needs measure. At the time of our study (data collected in 2008), the population was around 13 million. According to Grantham-McGregor et al. (2007), in 2004 Ecuador was together with Peru, Bolivia and Paraguay among the four countries in South-America where 20 to 40 percent of children under 5 years are disadvantaged. In the rest of South-America the percentage of disadvantaged children below age 5 is less than 20 percent. Paxson and Schady (2007) document a strong association in Ecuador between socio-economic status and children's early cognitive development. Schady (2011) shows that this association persists into primary school age and that early scores are strong predictors of later scores.

2.2 The interventions

The two programs that we examine in this paper are CIBV (for Centros Infantiles del Buen Vivir) and CNH (for Creciendo con Nuestros Hijos). The first is a program for child care centers, the second is a program for home visits. Both were publicly funded by the Child Development Fund (FODI), whose role and the way it selected providers we describe in Section 2.4. Both programs figure prominently in the report of Harris-Van Keuren and Rodríguez Gómez (2013), in which the authors present a comparative analysis of the guidelines of 19 programs for infants and toddlers under the age of 3 years in Latin America and the Caribbean.

One third of the children served by these programs attend a child care center, the other two thirds are exposed to home visits. Both programs have as their main objective to improve the early development of young children from poor families. The programs are provided at the community or neighborhood level. There is never more than one program per community/neighborhood. Children who are not served by one of the programs (the control groups) are normally looked after by their mother or another family member (grandparents or older sibling) or a neighbor.

Child care centers. The child care centers provide daycare, nutrition (breakfast and lunch) and educational services. It is full time: 52 weeks per year, 5 days per week, and 8 hours per day. Trained teachers work with groups of 8 to 10 children. There is no involvement of other adults. One center serves at most 45 children in the age range of 0 to 72 months.[1] The annual cost of the program amounts to US$ 488 per child. To put this amount into perspective, the average monthly income in our sample is US$ 300 per family or US$ 55 per capita.[2] FODI contributes at most 80% of the total costs of child care centers; 20% has to come from other sources. The first column of Table 1 shows the breakdown of the total costs of child care centers into various components; the main components are wages of the pedagogical and administrative staff and food for the children.

Table 1. Breakdown of the costs of the programs

Financial Structure	Child care centers	Home Visits
Wages of administrative staff	13.4%	33.2%
Initial costs	9.8%	0.5%
Wages of pedagogical staff	31.5%	45.4%
Training	1.5%	3.0%
Food	37.5%	Not provided
Monitoring and evaluation	0.7%	2.2%
Pedagogical material	1.0%	4.5%
Others	4.6%	11.2%

Source: FODI.

The child care centers work with a curriculum for preschool education which is designed by the Ecuadorian government for children between 0 to 72 months old. This curriculum includes competencies and activities for the four classical domains of early childhood development: i) physical/motor development (including gross and fine motor skills); ii) socio/emotional development (including the notion of self-concept, self-confidence and the ability to interact with the environment); iii) language development (including verbal and nonverbal communication and the ability to initiate a conversation),

[1]Primary education in Ecuador starts when children are between 6 and 7 years old. Enrollment in primary school is almost universal, but drops sharply at the secondary school level (see Oosterbeek et al., 2008).

[2]In 2000, Ecuador adopted the US dollar as its official currency.

and; iv) cognitive development (including abilities such as recognition, memory, imitation, problem solving and counting). For each domain, the curriculum lists competencies expected of children in a specific age range. Each competency is linked to indicators which are used to monitor children's progress. For each class (or meeting) an activity agenda is planned according to the curriculum.

According to the program guidelines, the child care centers are targeted to communities where mothers work or are likely to work. We discuss the channels through which the centers may affect children and mothers in Section 2.3.

Home visits. Through home visits, FODI attempts to stimulate children and to improve parents' attitudes, knowledge and behavior towards the development of their children. An important aim of the home visits is to teach mothers how to engage with their child in enriching activities, how to interact with their child in a nonaggressive way, how to create a responsive environment and how to prepare nutritional meals for their children. During a home visit, advice is communicated directly by explaining what, how, when and why an activity is performed. Activities are based on guidelines set by FODI. These guidelines are based on the same curriculum as the child care center intervention.

Children and their mother are treated individually when the child is younger than 3, and in groups of on average 22 children when the child is above 3. Home visits last 1 hour per week. The annual cost of this intervention is US$ 109 per child. FODI's contribution to home visits was up to 95% of this; only 5% has to come from other sources. The second column of Table 1 shows a breakdown of the costs of home visits into various components. Wages of teachers and administrative staff make up around 80% of the costs of this intervention.

Although there is no administrative record of the actual profile of the advisors working in the evaluated programs, the guidelines for the competitive process made no discrimination of gender among advisors. On the other hand, although it was not compulsory it was suggested that the advisors knew the locality as they were required to get to the community on a weekly basis. Moreover, they were also supposed to be in permanent contact with the community authorities in the rural area and keep them informed about their progress. FODI's application rules demand that instructors have at least a high school diploma and that they have at least one year of prior experience in the field of early childhood development. Instructors of awarded proposals received specific training about home visits during a period of three months.

According to the program guidelines, the home visits are targeted to communities where mothers are more likely to stay at home.

Table 2 summarizes the main characteristics of the two programs.

Table 2. Main characteristics of interventions

Program characteristic	Child care centers (CDI)	Home Visits (CNH)
Child Care services	Center based intervention consisting of nursery and pedagogical stimulation	No child care provided
Parental support	No parental support provided	Weekly session with parents and children at home for children 0 to 36 months and at a community center for older children
Nutritional services	Breakfast and lunch	Nutritional education for parents
Age group served	0 to 72 months	0 to 72 months
Operating schedule	52 weeks per year, 5 days per week, 8 hours per day	12 months per year, 1 day per week, 1 hour per visit
Residential Area coverage	Urban and rural	Urban and rural
Target population	Low income children in poor areas with working or likely to work mothers	Low income children in poor areas with non-working mothers
Children/instructor ratio	8 to 10 children. Maximum 45 children per center	Individual intervention at home 22 children per educator for children >3
Level of education of instructor	High school degree	High school degree; bachelor degree
Annual Cost per child (2006)	US$487.84	US$109.26

Quality and implementation. Harris-Van Keuren and Rodríguez Gómez (2013) give both interventions a score of 0.5 on a scale from 0 to 1. The meaning of this score is that the curricula of the two interventions meet the ideal indicators for 50%. The two programs can therefore not be regarded as showcase interventions. A score of 0.5 is, however, equal to the mean score of the 19 programs in Latin America and the Caribbean that the authors have assessed. This suggests that the programs considered in our evaluation are representative for this type of intervention in the region.

In the child care program, the curriculum concentrates 34% on physical/motor skills (versus 43% for the average of similar programs in the region), 24% on cognitive development (versus 17%), 25% on language (versus 23%), and 17% on social/emotional skills. Compared to the regional average, the Ecuadorian child care program therefore puts somewhat more emphasis on cognition and language and somewhat less on motor skills.

For the home visit program, the curriculum concentrates 34% on physical/motor development, 25% on cognitive development, 24% on language, and 17% on social and emotional development. The regional averages for comparable home visit programs are 20%, 21%, 26% and 33% respectively. Relative to the regional average, the Ecuadorian home visit program puts more emphasis on motor, cognitive and language development and less on social and emotional development.

FODI's application rules require that the home visit interventions started as soon as the contract was signed while the child care interventions were supposed to start at most 30 days after the contract had been signed. The latter is also shown by the data that we collected on length of treatment. Treated children in our sample had an average exposure to treatment of 21 months and this exposure is the same for treated children in either home visits or child care centers.

2.3 Mechanisms and expected effects

The two programs may affect children and mothers through different channels. To organize the empirical analysis, we use a simple framework in which child development is produced by the quantity and the quality of the care time the child receives and by the material inputs available at home, and in which the mother allocates her time between child care, market work and other activities (cf. Attanasio, 2026). Within this framework we state, for each program, the outcomes for which we expect effects and the direction of these effects.

Child care centers substitute center-based care for maternal or family care during the day and provide two meals. We therefore expect the centers to raise mothers' labor supply and family income; this is the most direct channel (cf. Berlinski and Galiani, 2007;

Berlinski et al., 2011). For children, the expected effect on development is positive if the quality of care in the centers exceeds the quality of the care they replace, and can be zero or negative otherwise (Baker et al., 2008; Havnes and Mogstad, 2011). The meals provided may improve children's nutrition and health. The expected effect on mothers' psychological wellbeing is ambiguous: additional income and relief from daytime care may improve it, while combining work with the remaining care obligations may worsen it.

Home visits do not provide care; they aim to change the quality of the mother's interaction with the child. We therefore expect the most direct effects on children's development and on parenting practices, and, through the nutrition component, possibly on children's health. The expected effect on mothers' labor supply is ambiguous. A weekly one-hour visit does not mechanically constrain the mother's time. Yet the program may raise the perceived return to maternal time with the child, which raises the opportunity cost of working; it may also leave labor supply unchanged, or raise it if care routines become more efficient. We treat the sign of this effect as an empirical question. The expected effect on mothers' wellbeing is positive if the quality of mother-child interactions improves.

Two implications of this framework guide the interpretation of the results. First, the primary outcomes differ across programs: maternal labor supply for child care centers, child development and parenting for home visits. Second, because the programs were proposed for different communities and differ in intensity (8 hours per day versus 1 hour per week), delivery agent and setting, differences between the two sets of estimates may reflect any of these dimensions. We therefore evaluate the two programs in parallel and treat the cross-program comparison as descriptive (Section 6.5). We also note that the targeting described in the program guidelines is not visible in our data: among unfunded proposals, mothers in home-visit communities work more than mothers in child-care communities (Section 6.5).

2.4 Selection process

At the time of our study, early childhood development programs in Ecuador were offered by both private and public providers. Private provision was small and mainly targeted towards middle and high income families, while public provision primarily served low income families, targeting children in rural and poor urban areas. There were three main public providers; the largest was FODI ("Fondo de Desarrollo Infantil").[3] FODI started

[3]The other two large public providers were INNFA (for "Instituto del Niño y la Familia") and ORI (for "Operación Rescate Infantil"). In July 2008, all three were merged into the Instituto de la Niñez y la Familia (INFA) by Executive Decree No. 1170. INFA was itself absorbed into the Ministry of Economic and Social Inclusion (MIES) in 2012 (Executive Decree No. 1356).

in 2005 and, at the time of our study, served around 300,000 children from poor families all around Ecuador.

FODI does not run its own centers but subsidizes non-profit suppliers of early childhood development centers. FODI allocates its budget through a contest. It organized such contests in 2005, in 2006 and in 2008. In this paper we use data from applicants (winners and losers) to the contest of 2006. This contest was organized to expand services to poor unserved neighborhoods or communities. The available budget for this contest was US$ 12 million. 240 organizations submitted a proposal, 95 of which were awarded for a total coverage of 60,000 children. Awarded proposals initially receive funding for a two year period, which is normally renewed afterward.

One proposal could, and typically did, involve the opening of centers in more than one community. A given community could, however, not be included in more than one proposal. All centers in a given proposal offer the same type of program (either home visits or child care centers) and pertain to only one province.

In the 2006-contest, a proposal was only considered for funding if: (i) it included a list with the names of the children it would serve if awarded, (ii) the area was not yet served by a public provider, and (iii) the proposal fulfilled certain standards regarding infrastructure and educators' skills. Proposals meeting these requirements were given a score based on five criteria (behind each criterion, the maximum number of points that can be earned for it): socioeconomic characteristics of the neighborhood (180); coherence of the proposal (130); share of outside funding (30); quality of personnel (150); financial aspects (110).

Adding the maximum scores per element gives an overall maximum score of 600. FODI then allocated its available budget for the second contest by funding proposals from high to low until the budget was exhausted. In practice FODI spent the last dollar of the available budget on a proposal that received an overall score of 425.[4] This last proposal offers home visiting programs, but the same cutoff score then applies to proposals for child care centers. All proposals for child care centers with a score above 425 received funding, while no proposal for child care centers with a score below 425 received funding. We will exploit the funding threshold in a regression discontinuity design. The threshold could not be anticipated by the applicants or the organizers, reducing the likelihood of manipulation.[5] The discontinuity applies at the level of proposals and not at the level of children listed in a proposal. There is no ranking of individual children within or across proposals.

[4]Beforehand, a score of 400 was set as the minimum required quality standard. Proposals with a score below 400 would not qualify for funding even if the available budget was not exhausted.

[5]In Section 5 we show a density plot of the number of proposals by score. There is indeed no sign of manipulation and McCrary's density test fails to reject the null of no difference in density estimates just to the left and just to the right.

The two interventions are still in operation today, now under Ecuador's "Misión Ternura" early childhood strategy and administered by what is currently the Ministry of Human Development. What changed is therefore not the programs but the way they are funded: the FODI proposal-scoring mechanism that we exploit was discontinued through the mergers described above. This makes the regression discontinuity we study a historical episode, while the policy question it speaks to remains current.

3 Empirical approach

3.1 Design and estimation

We are interested in the causal impact of exposure to child care centers and home visits on outcomes of children and their mothers. Naive OLS estimates are likely to be biased by selection: programs target disadvantaged communities, so without intervention treated children would probably have worse outcomes than the untreated. Conversely, parents may enroll children they perceive as particularly able.

The way in which FODI allocated its budget provides a regression discontinuity (RD) design (cf. Lee and Lemieux, 2010; Cattaneo and Titiunik, 2022). For each program type, proposals with a score at or above the funding threshold received support while those below did not. We instrument actual program exposure with the indicator $Z_i = \mathbf{1}[s_i \geq s_0]$, where s_i is the proposal score and $s_0 = 425$ is the cutoff. Compliance on the funded side is 97–99%, so the design is nearly sharp. The causal impact of home visits is identified from:

$$Y_i = \alpha_{HV} + \delta_{HV} HV_i + f_{HV}(s_i) + X_i \beta_{HV} + \varepsilon_{HVi} \tag{1}$$

where Y_i is an outcome, $HV_i = 1$ if the child was exposed to home visits and $HV_i = 0$ for children in the home-visit comparison group, and $f_{HV}(s)$ controls flexibly for the forcing variable. Analogously, the causal impact of child care centers is identified from:

$$Y_i = \alpha_{CC} + \delta_{CC} CC_i + f_{CC}(s_i) + X_i \beta_{CC} + \varepsilon_{CCi} \tag{2}$$

where $CC_i = 1$ if the child was enrolled in a child care center and $CC_i = 0$ for the child care comparison group. HV_i and CC_i are instrumented by Z_i. The parameters of interest are δ_{HV} and δ_{CC}, estimated on entirely separate samples: δ_{HV} uses children listed in home-visit proposals and δ_{CC} uses children listed in child care center proposals, so the same instrument is never used to identify two parameters simultaneously.

We estimate both equations by local-linear (p = 1) 2SLS with a triangular kernel and an MSE-optimal data-driven bandwidth chosen separately for each outcome and program using `rdrobust` (Calonico et al., 2014). The function $f(s)$ is approximated by a linear polynomial on each side of the cutoff within the bandwidth window; no global functional form is imposed. We report three specifications: (1) local linear without covariates; (2) local linear with covariate adjustment (Calonico et al., 2019) using MSE-optimal bandwidth; and (3) local linear with covariate adjustment using a coverage-error-rate (CER) optimal bandwidth, which is smaller than the MSE-optimal bandwidth and is chosen to minimize the error in confidence interval coverage rather than the bias–variance trade-off of the point estimate. Covariates include child sex, household size, urban status, welfare receipt, mother's age, schooling of mother and household head, mother's vocabulary score, parenting-practice dummies, ethnicity, household composition, and mother's literacy. Standard errors are clustered at the level of project $\times$ parish (98 clusters). All point estimates are bias-corrected and confidence intervals are bias-robust.

3.2 Inference with a discrete, proposal-level forcing variable

A feature of our design that complicates inference is that the funding score takes only 34 distinct values (one per proposal), because the score is constant within a proposal and there are 34 proposals in the vicinity of the cutoff. This creates two distinct problems. First, the effective number of independent assignment units is 34, not 2,707, so analytic clustered standard errors may under-cover. Second, the support of the forcing variable is discrete, so the data within any bandwidth cannot reveal the curvature of the conditional expectation function between mass points, and conventional bias-corrected confidence intervals lose their justification (Kolesár and Rothe, 2018). Appendix Table B1 documents the effective support: for each outcome it reports the bandwidth, the number of proposals on each side of the cutoff within the bandwidth, and the effective number of children. The number of proposals within the bandwidth ranges from 2 to 7 below the cutoff and from 3 to 6 above. The mass points nearest to the cutoff lie at -0.7 and $+0.1$ normalized score units in the home-visit sample and at -2.7 and $+0.7$ in the child-care sample, so the local-linear fit extrapolates over a small gap at the boundary itself.

Our main tables cluster standard errors at the level of project $\times$ parish (98 cells). Because treatment is assigned at the proposal level, the design-based view of clustering (Abadie et al., 2023) points to the proposal as the natural cluster. Appendix Table B2 reports the main estimates with standard errors clustered on the 34 proposals; the resulting p-values are similar to, and often smaller than, the project $\times$ parish ones, so the choice between the two levels does not drive our conclusions. The relevant concern is rather that any analytic cluster-robust variance estimator with 34 clusters may under-cover. We address the two problems with three complementary procedures.

First, we compute wild-cluster-bootstrap p-values over the 34 proposals (Cameron et al., 2008). The bootstrap addresses the small number of assignment units. It maintains the local-linear specification and therefore cannot address the discrete support; resampling proposals does not create score values that are not in the data.

Second, we report bias-aware honest confidence intervals (Kolesár and Rothe, 2018; Armstrong and Kolesár, 2020). These remain valid with a discrete running variable, but they require an explicit bound M on the second derivative of the conditional expectation function. This bound cannot be chosen in a data-driven way without invalidating the coverage guarantee, so we report the rule-of-thumb value of M for every outcome and show how the intervals change when M is doubled and quadrupled. Because our first stage is 0.97–0.99, the reduced-form and fuzzy parameters nearly coincide; we nonetheless also report fuzzy bias-aware intervals following Noack and Rothe (2024), which account for the estimated first stage.

Third, we report finite-sample randomization inference in a narrow window around the cutoff (Cattaneo et al., 2015). This approach assumes that proposals within the window are as good as randomly assigned to the two sides of the cutoff. Since treatment is assigned at the proposal level, the appropriate permutation unit is the proposal. This requirement is restrictive in our data: the ± 5 window contains only three proposals in the home-visit sample and two in the child-care sample, so a proposal-level permutation test cannot produce a p-value below one third in that window. We report randomization p-values for a range of windows, at both the child and the proposal level, together with the number of proposals in each window (Table 12). As we show below, the proposal-level tests are uninformative in this design, and we therefore do not count randomization inference as independent support for any finding.

The three procedures address different problems and rest on different assumptions, and none dominates the others ex ante. When we label results, we call an effect robust if its honest confidence interval excludes zero and this conclusion survives the reported variations in the smoothness bound; we call an effect suggestive if it is significant under conventional clustered inference but does not meet this standard.

3.3 Comparing the two programs

Comparing δ_{HV} with δ_{CC} requires care. The two estimates come from different proposal samples and populations, and the interventions differ in intensity, delivery agent and setting. Even within the bandwidth window, HV and CC applicants differ on observable characteristics: home-visit proposals were more likely to be urban and served non-indigenous populations at higher rates (Table 9). When we place the two sets of estimates side by side we use the covariate-adjusted specifications (specification 2), which absorb these observable differences, and we compute the standard error of the difference $\hat{\delta}_{HV} - \hat{\delta}_{CC}$ as $\sqrt{SE_{HV}^2 + SE_{CC}^2}$, which is valid because the two samples are drawn from independent communities. Covariate adjustment does not, however, account for differences in unobservables, and no adjustment can separate the effect of program type from the effects of intensity, delivery and population. We therefore present the comparison in Section 6.5 as descriptive evidence, not as an estimate of the causal effect of offering one program rather than the other.

4 Data

A total of 113 submitted proposals passed the minimum quality requirement of 400 points. Because data collection is expensive and available resources for this project limited, we focused data collection on proposals with a score relatively close to the funding threshold. For the appreciation of the results reported in this paper, it is important to consider that we thus start with a sample that can be regarded as a discontinuity sample (e.g. Angrist and Lavy, 1999). We subsequently selected a random sample of the centers that were included in these proposals. A proposal can include multiple centers, but only for the same type of child care program. The sampling was stratified by: assigned treatment status, type of program (home visits or child care centers), urban or rural area, center size and province (a proposal only serves communities in the same province).

Within each center, we selected a random sample from the children whose names were on the list attached to the proposal. Notice that the lists with names of the children that would be served are vital for our design. Without those lists it is impossible to know which children would have been treated by the providers that did not receive funding.

The final sample consists of 2,707 children in 99 centers; 38 providing child care centers and 61 home visits. Table 3 shows the numbers of observations (children and centers) in our final sample by type of program, age group and treatment eligibility.[6] We present a breakdown by age because some cognitive and motor tests are only validated for children older than 36 months while others are only validated for children younger than 60 months.

[6]As we will show in Section 5 treatment eligibility and actual exposure to treatment are very highly correlated.

Table 3. Number of children by program, eligibility and age (number of centers in parentheses)

Program	Age	Non-eligible	Eligible	Total
Home visits	all	830 (28)	988 (33)	1818 (61)
	age >36 months	739 (27)	794 (33)	1533 (60)
	age <60 months	465 (27)	704 (33)	1169 (60)
Child care centers	all	411 (12)	478 (26)	889 (38)
	age >36 months	371 (11)	421 (26)	792 (37)
	age <60 months	194 (12)	318 (26)	512 (38)

Teams of data collectors visited the homes of all children included in the final sample. They collected data through interviews with the mothers and through tests and measurements. Data were collected between September and December of 2008. At the moment of data collection, treated children in our sample had an average exposure to treatment of 21 months. This is the same for child care centers and home visits.

Item non-response is small. Among sampled children, the Nelson-Ortiz test was completed for 98 percent, the anthropometric measurements for 96 to 98 percent, and the tests administered to children older than 36 months (Peabody, Woodcock, Pegboard) for 96 to 97 percent of the eligible children. The exception is the blood sample used to measure anemia, which was obtained for 86 percent of the children; the anemia results should be read with this in mind. The survey did not produce design weights, and none of the estimates below are weighted. The estimates therefore describe the sampled children of applicant proposals near the funding threshold. This is the population of interest in a regression discontinuity design, which in any case identifies effects local to the cutoff.

We used standard and validated test instruments to measure the cognitive, motor and social-emotional development of children. Some tests are specific for children older than 36 months. These are the Spanish versions of the Peabody Picture Vocabulary test which measures receptive vocabulary (language), the Woodcock-Johnson-Munoz test which measures long term memory, and the Pegboard test which measures fine motor skills. For all children from 0 to 60 months old, we use the Nelson-Ortiz test which measures four dimensions of child development: language skill, gross motor skill, fine motor skill and social behavior.

Test scores are standardized by age. The Peabody, Woodcock and Pegboard instruments produce age-normed scores, which we normalize to mean zero and standard deviation one; impact estimates for these outcomes are thus expressed in standard deviation units. The Nelson-Ortiz instrument (the Escala Abreviada de Desarrollo) scores each child on the items for the child's age band and classifies the total score into four categories using the scale's published norm cutoffs, which are defined for ten age bands between 1 and 72 months. Our binary outcome equals one if the child falls in one of the two categories above the norm-population mean for the child's age band, and zero otherwise. The reference group is thus the external norm population of the scale, not our sample, so the outcome definition does not depend on the age composition of our sample. Appendix A describes the instruments, the classification rules, and the completion rates in more detail.

To measure children's health we use height for age and weight for age. Height for age is an indicator for long-term health outcomes, while weight for age more reflects short-term health conditions. We also took blood samples to measure the hemoglobin levels of the children to detect iron deficiency anemia. For the mothers we measure the following outcomes: the Center for Epidemiological Studies Depression (CES-D) scale to measure depression and psychological stress, the Home Observation for Measurement of the Environment (HOME) scale to measure responsiveness to children, and variables related to the labor market such as participation, working hours and income. The CES-D covers the main symptoms of depression and is derived from five validated depression scales. The score on the HOME test is based on the interviewer's evaluation of the mother's attitudes and behavior towards the child during the interview. We converted respondents' scores on the CES-D test and the HOME test to mean zero and standard deviation one, so that again impact estimates are measured in standard deviation units.

5 First stage and identifying assumptions

In this section we will first show that whether a proposal is above or below the funding threshold almost perfectly determines whether a child that is on the list of a proposal, is exposed to treatment. We then discuss the identifying assumption and provide evidence in support of it.

All proposals with a score above the threshold received funding from FODI and all proposed programs had been implemented at the moment of data collection. Likewise, none of the proposals with a score below the threshold received funding from FODI and none of these proposed programs have been implemented. At the proposal or program level, the allocation of FODI's budget thus represents a sharp regression discontinuity design; the score assigned to the proposal perfectly determines whether the proposal receives funding and whether the proposed program is implemented.

The sharp design at the proposal or program level translates into an almost sharp design at the level of children. Just a few children included in a proposal that received funding did not participate in the program, and also just a few children included in a proposal that did not receive funding participated in an(other) early childhood program.[7]

[7] Two children did not participate in a home visit program while they should, 9 children did not partici-

The almost perfect compliance with the assigned treatment status results in first stage estimates close to one. Figure 1 plots the near-perfect jump in program take-up at the funding cutoff for both programs; Table 4 quantifies this with the same local-linear specifications as the main results tables.

Table 4. First stage regressions

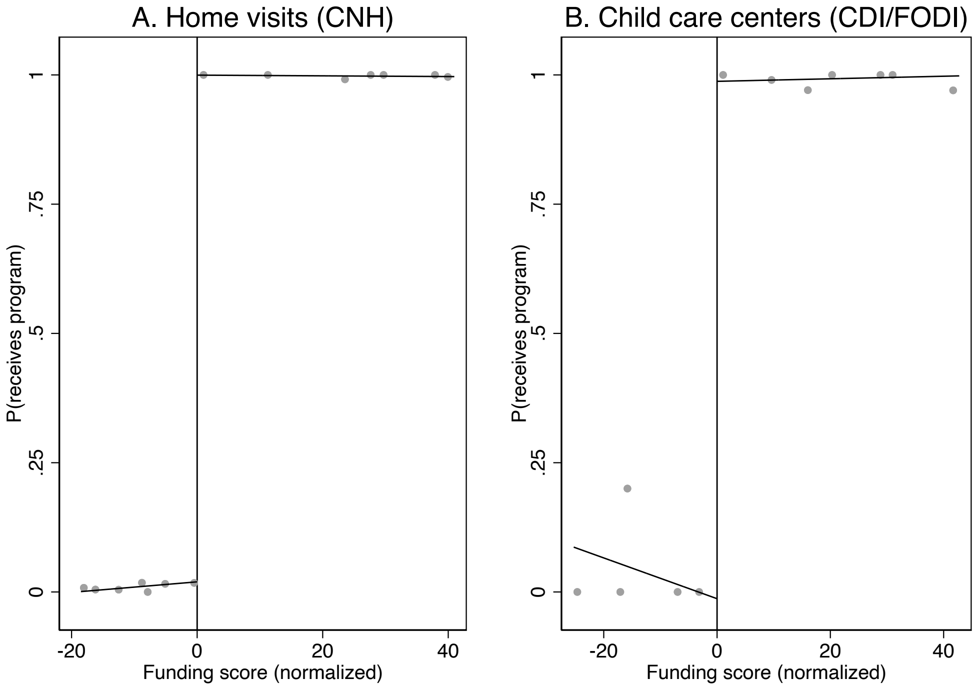
	(1) LL, MSE	(2) LL, MSE+X	(3) LL, CER+X
A. Home visits			
Above threshold (Z)	0.981***	0.976***	0.976***
	(0.011)	(0.007)	(0.007)
Effective N	1174	1147	1029
Bandwidth h	24.7	23.8	19.4
B. Child care centers			
Above threshold (Z)	1.010***	1.089***	1.011***
	(0.030)	(0.031)	(0.035)
Effective N	308	418	305
Bandwidth h	14.3	15.9	13.2

Notes: Each cell reports the bias-corrected RD estimate of the effect of being above the funding threshold on actual program take-up, with robust standard error in parentheses, obtained with `rdrobust` (Calonico et al., 2014). Specifications match those of Tables 13–16: (1) local linear (p=1), MSE-optimal bandwidth, no covariates; (2) local linear, MSE-optimal, with covariate adjustment (Calonico et al., 2019); (3) local linear, CER-optimal bandwidth, with covariates. Standard errors clustered on project × parish. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

As mentioned before, proposals with a score above 425 (on a scale from 0 to 600) received funding and proposals with a lower score did not. Table 5 shows average values of the scores for proposed programs above and below the threshold in our sample, separately for home visits and for child care centers. The table also shows the average scores the proposals received on each of the five components of the total score. Because the score is assigned at the proposal level, we compute these averages over the 34 proposals rather than over centers; with so few proposals the tests have limited power. For home visits, the only component that differs significantly between funded and unfunded proposals is the quality of staff (column (3)); the differences in the other components, including the socio-economic background of the communities served, are not significant. For child care centers, the quality of staff and the coherence of the proposal differ between funded and unfunded proposals, though only marginally given the small number of proposals (column (6)); the remaining components do not differ significantly. The final column compares home-visit and child care center proposals. The point estimates suggest that

pate in a child care center program while they should, 7 children participated in a home visit program while they should not, and 5 children participated in a child care center while they should not.

Figure 1. First stage: program take-up by funding score



Notes: Binned means of treatment take-up against the normalized funding score, estimated by `rdplot` (Calonico et al., 2014). Left panel: home visits (CNH); right panel: child care centers (CDI/FODI). Take-up jumps from near zero to near one at the cutoff in both programs, confirming a near-sharp design. Compliance rates are 97–99% on the funded side; the quantitative first-stage estimates are in Table 4.

home visits are proposed for somewhat better-off communities (they collect fewer points for SES), but with only 34 proposals this difference is not statistically significant; the components that do differ across programs are co-funding and the community share of child labor. Total scores do not differ significantly between the two programs, so by this grading system the two interventions are of comparable overall quality. The more salient differences between the populations served, in urbanicity and ethnicity, are documented when we compare the two programs (Table 9), and motivate the covariate adjustment used there.

Table 5. Components of score

Variable	Home visits (HV)			Child care centers (CC)			HV vs CC
	$s < 425$	$s \geqslant 425$	p	$s < 425$	$s \geqslant 425$	p	p
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Total score	415.69	452.96	0.000	412.10	445.26	0.000	0.383
	(6.24)	(14.39)		(8.88)	(12.96)		
SES	174.29	173.46	0.800	178.00	173.89	0.110	0.441
	(6.73)	(6.89)		(2.74)	(4.86)		
Co-funding	25.33	26.43	0.739	20.40	22.67	0.214	0.041
	(7.98)	(6.43)		(0.89)	(3.74)		
Quality of staff	22.65	48.67	0.000	23.71	45.39	0.054	0.772
	(12.83)	(11.97)		(17.53)	(18.55)		
Coherence	105.97	110.56	0.368	102.43	113.79	0.094	0.843
	(11.35)	(10.20)		(15.07)	(8.67)		
Financial aspects	87.46	93.84	0.213	87.56	89.52	0.750	0.457
	(8.76)	(11.32)		(11.10)	(10.68)		
N	7	13		5	9		34

Notes: Average values of the total funding score and its five sub-components for proposals below ($s < 425$) and above ($s \geqslant 425$) the funding threshold, separately for home visit (HV) and child care center (CC) proposals. The unit of observation is the proposal, which is the level at which the score is assigned and treatment is determined: there are 34 proposals (20 HV, 14 CC), each with a single score. Standard deviations in parentheses. Columns (3) and (6) report p-values from a t-test for the difference in means between below- and above-threshold proposals; column (7) the p-value for the difference in means between HV and CC proposals (pooling below and above threshold). Given the small number of proposals these tests have limited power. The funding score ranges from 0 to 600 and is composed of five criteria: socioeconomic status of the target community (SES), co-funding commitment, quality of proposed staff, coherence of the project plan, and financial soundness.

The identifying assumption in the regression discontinuity design is that conditional on a smooth function of the underlying score and observables included in the analysis, there are no systematic unobserved differences between observations just below the threshold and observations just above the threshold. While this assumption cannot be tested, we can test whether observations above and below the threshold are not systematically different in terms of their observable characteristics.

Tables 6 and 7 show the mean values of observable characteristics. The first for variables measured at the community level, the second for variables measured at the level of children and their mothers. The tables show the average values for community and background variables separately for observations below and above the funding threshold and separately for the two programs. They also report p-values for the difference in means between below- and above-threshold groups. For the community-level variables in Table 6 we test at the proposal level, the level at which the score is assigned and treatment is determined (34 proposals), using a simple difference in means; for the child- and mother-level variables in Table 7 we test at the individual level, conditioning on a second-order polynomial of the forcing variable. The p-values in columns (3) and (6) of Table 6 indicate that, for both programs, funded and unfunded communities are not significantly different on the community characteristics. The only difference significant at conventional levels is the school supply per 1,000 inhabitants for home visits ($p = 0.05$), and even there the point difference is modest. (The apparent imbalances in level of schooling and access to water that arise when the test is run at the center level do not survive once the proposal is used as the unit of analysis.)

Table 6. Differences in community characteristics, by eligibility status

Variable	Home visits (HV)			Child care centers (CC)			HV vs CC
	$s < 425$	$s \geqslant 425$	p	$s < 425$	$s \geqslant 425$	p	p
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Poverty (share)	0.60	0.65	0.293	0.69	0.58	0.336	0.760
	(0.14)	(0.08)		(0.19)	(0.18)		
School enrollment (share)	0.93	0.92	0.495	0.89	0.92	0.175	0.348
	(0.03)	(0.03)		(0.04)	(0.03)		
High school enrollment (share)	0.68	0.65	0.480	0.64	0.67	0.695	0.923
	(0.07)	(0.10)		(0.12)	(0.12)		
Level schooling (years)	8.47	7.32	0.120	6.26	7.78	0.287	0.493
	(0.67)	(1.78)		(1.71)	(2.75)		
Child labor (share)	0.02	0.03	0.520	0.06	0.03	0.097	0.022
	(0.01)	(0.01)		(0.04)	(0.02)		
Access water (share)	0.39	0.29	0.218	0.18	0.38	0.169	0.776
	(0.15)	(0.18)		(0.16)	(0.27)		
Access sanitation (share)	0.77	0.80	0.512	0.70	0.74	0.572	0.051
	(0.07)	(0.09)		(0.11)	(0.13)		
Outpatient health centers (per 1000)	2.63	3.19	0.445	2.77	3.17	0.471	0.941
	(1.17)	(1.67)		(1.00)	(0.96)		
Inpatient health centers (per 1000)	0.34	0.27	0.504	0.52	0.45	0.616	0.027
	(0.21)	(0.23)		(0.34)	(0.18)		
School supply (per 1000)	1.62	2.70	0.046	2.08	1.98	0.881	0.457
	(0.74)	(1.21)		(1.11)	(1.30)		
High school supply (per 1000)	0.05	0.05	0.754	0.05	0.04	0.793	0.744
	(0.02)	(0.03)		(0.02)	(0.02)		
N	7	13		5	9		34

Notes: Mean values of community-level characteristics for proposals below and above the 425-point funding threshold, separately for home visits (HV) and child care centers (CC). The unit of observation is the proposal (34 in total: 20 HV, 14 CC); community-level variables are averaged to the proposal. Standard deviations in parentheses. p-values in columns (3) and (6) are from t-tests for equality of means between below- and above-threshold proposals within each program; column (7) reports the p-value for the difference in means between all HV and all CC proposals. With so few proposals these tests have limited power.

Table 7. Differences in characteristics of children and mothers, by eligibility status

Variable	Home visits (HV)			Child care centers (CC)			HV vs CC
	$s < 425$	$s \geqslant 425$	p	$s < 425$	$s \geqslant 425$	p	p
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Boy (dummy)	0.52	0.51	0.050	0.50	0.52	0.274	0.743
	(0.50)	(0.50)		(0.50)	(0.50)		
Age (months)	55.8	49.6	0.000	58.3	52.0	0.002	0.055
	(14.4)	(12.9)		(14.3)	(12.4)		
Household size (persons)	4.87	4.85	0.437	5.70	5.25	0.245	0.026
	(1.71)	(1.71)		(1.97)	(2.00)		
Urban (dummy)	0.70	0.66	0.934	0.49	0.40	0.802	0.088
	(0.46)	(0.48)		(0.50)	(0.49)		
Cash transfer (dummy)	0.58	0.55	0.388	0.61	0.58	0.036	0.460
	(0.50)	(0.50)		(0.49)	(0.49)		
Mother's age (years)	30.7	30.5	0.847	31.6	30.6	0.109	0.283
	(8.56)	(8.12)		(8.46)	(7.96)		
Schooling mother (years)	7.24	7.61	0.421	5.38	5.96	0.375	0.000
	(3.66)	(4.13)		(3.55)	(3.56)		
Schooling head (years)	6.66	6.72	0.968	5.51	5.74	0.854	0.006
	(3.87)	(5.36)		(3.50)	(3.37)		
Language score mother	72.1	70.5	0.994	59.78	62.2	0.398	0.016
	(23.9)	(26.2)		(26.0)	(25.1)		
Father present (dummy)	0.72	0.81	0.436	0.83	0.77	0.154	0.324
	(0.45)	(0.39)		(0.38)	(0.42)		
Mother present (dummy)	0.95	0.96	0.721	0.95	0.96	0.940	0.734
	(0.22)	(0.21)		(0.23)	(0.19)		
N	830	988		411	478		2707

Notes: Mean values of child and mother characteristics for children below and above the 425-point funding threshold, separately for home visits (HV) and child care centers (CC). Observations are at the child level. Standard deviations in parentheses. p-values in columns (3) and (6) are for the below-vs-above difference within each program, conditioning on a second-order polynomial of the forcing variable, with standard errors clustered on project × parish. Column (7) reports the p-value for the difference in means between all HV and all CC children.

The p-values in columns (3) and (6) of Table 7 indicate that the characteristics of children eligible for treatment are in most cases not significantly different from the characteristics of children not eligible for treatment. There is only a systematic difference in age.[8] In the samples of both programs eligible children are significantly younger than non-eligible children; the age gap is about 6 months. Part of this difference can be attributed to eligible children being interviewed earlier than non-eligible children. But even after we correct for that, a significant difference remains. We have no explanation for this.

To assess whether the difference in age between treatment and control groups can drive

[8]The differences between eligible and non-eligible children are also statistically significant for gender in the home visits program and for receipt of the cash transfer in the child care center program. The means of the these variables are, however, almost identical for eligible and non-eligible children.

our results, we regressed each outcome variable on age using data from the control groups only. We did that for the two control groups together and separately. For most outcomes there is no significant relation with the age of the child. There are three exceptions: fine motor skills for children older than 36 months, anemia and non-responsiveness of the mother. In all three cases, the signs of these relationships are, however, the opposite of what would explain our findings. For instance, the relation between age and fine motor skills is positive. In the absence of a treatment effect, this would imply that eligible children do worse than non-eligible children. This implies that estimated treatment effects that are not, or insufficiently, corrected for the age pattern, are biased downwards. Similar reasonings apply to the effects of the treatments on the other two outcomes. We will report results from specifications i) that do not control for age (other than that some outcome variables are standardized by age); ii) that only control for age (and the running variable); iii) that control for age along with other covariates.[9]

The age imbalance deserves particular attention for the Nelson-Ortiz overall development outcome, which is our most robust result and which is defined relative to age-band norms. We therefore re-estimate this effect within age ranges that are common to both sides of the cutoff. Table 8 reports the results. In the home-visit sample, restricting the sample to ages 30–59 months (the overlap of the 5th to 95th percentiles of the age distributions on the two sides) removes the age discontinuity entirely (a difference of -0.8 months, $p = 0.48$), while the estimated effect on overall development is 0.37 ($p < 0.001$), slightly larger than the full-sample estimate of 0.30. The same holds for ages 36–60. The effect on overall development is therefore not an artifact of the younger age of eligible children.

The last column reports p-values for differences in characteristics between children whose name appeared on the list of a proposal for home visits and children whose name appeared on the list of a proposal for a child care center. These cross-program comparisons are relevant for interpreting the RD estimates side by side. The results show that children listed for child care centers are significantly older than children listed for home visits. There also is a difference in social background between children listed for home visits and children listed for child care centers. This difference shows up in the schooling levels of the mother and of the household head, mother's language score, and household size. As we already saw in Table 5, home visits serve children from relatively better-off families.

Because of the differences between the groups targeted by the two interventions we check whether the populations of HV and CC applicants are comparable *within* the MSE bandwidth window. Table 9 reports covariate means for HV and CC applicants whose

[9] We also redid all analyses including higher order terms of age. This does not change any of the results.

Table 8. Effect on overall development within common age ranges

Age range (months)	N	RD estimate (LL, $+X$)	(s.e.)	Age discontinuity months	p
Panel A. Home visits					
All ages 0–60	1200	0.300***	(0.086)	-3.2	0.000
Common support 30–59	1080	0.368***	(0.097)	-0.8	0.476
Ages 24–60	1200	0.300***	(0.086)	-3.2	0.000
Ages 36–60	947	0.305***	(0.098)	0.4	0.587
Ages 0–36	279	0.329***	(0.084)	-2.4	0.001
Panel B. Child care centers					
All ages 0–60	546	-0.019	(0.054)	10.1	0.000
Common support 30–59	491	-0.010	(0.055)	8.3	0.000
Ages 24–60	546	-0.019	(0.054)	10.1	0.000
Ages 36–60	460	-0.014	(0.056)	5.1	0.000
Ages 0–36	95	1.031***	(0.165)	0.1	0.874

Notes: Each row re-estimates the effect on the Nelson-Ortiz overall development score (the column (2) specification of Tables 13 and 14) on the sample restricted to the indicated age range. The common-support range is the overlap of the 5th to 95th percentiles of the age distributions below and above the cutoff. The last two columns report the RD estimate of the discontinuity in age itself within the restricted sample and its p-value; a small and insignificant age discontinuity indicates that the age imbalance is absent in that range. The child-care estimate for ages 0–36 rests on 95 children and should be discounted. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

proposal score falls within ± 15 normalized score units of the cutoff, along with the difference and its p-value. Even within this narrow window some differences remain: HV applicants are more likely to be urban (75 % vs. 36 %) and non-indigenous (67 % vs. 34 %), and have somewhat more educated mothers and household heads. The joint F-test rejects balance at the 5 % level. These differences reflect the program designs: home visits can operate in urban settings and require only a visiting worker, while child care centers require a physical facility and are more often located in rural or indigenous communities.

Table 9. Covariate balance: home visits vs. child care centers within ±15 score units of the cutoff

	HV	CC	Difference	p-value
Child age (months)	53.9	56.2	-2.3	0.354
Male	0.526	0.485	0.041	0.227
Children < 5 in household	1.7	2.0	-0.3**	0.032
Urban	0.753	0.361	0.392	0.118
Welfare transfer	0.556	0.590	-0.034	0.660
Mother age (years)	30.2	31.1	-0.9	0.225
Mother schooling (years)	7.4	5.6	1.7*	0.073
Head schooling (years)	6.6	5.7	0.9**	0.044
Mother vocab. score	71.7	58.4	13.3*	0.093
Parenting index 1	0.780	0.816	-0.036	0.402
Parenting index 2	0.964	0.944	0.020	0.206
Indigenous	0.328	0.656	-0.328***	0.004
Adolescents in household	1.1	1.7	-0.5	0.172
Adults in household	1.9	2.0	-0.1	0.323
Elderly in household	0.0	0.0	-0.0	0.350
Mother illiterate	0.072	0.167	-0.095	0.244
Joint F-test (16 covariates)			$F = 2.24$**	0.022

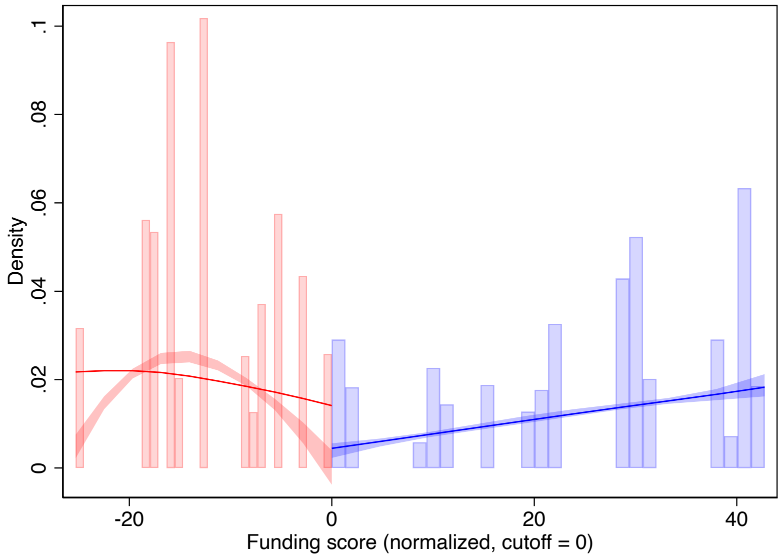
Notes: Sample restricted to children whose proposal score falls within ±15 normalized score units of the funding cutoff. Means are child-level; differences estimated by OLS with standard errors clustered on project × parish. Stars on the difference and the joint F-test denote significance: $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Because observable differences persist near the cutoff, a direct comparison of the two reduced-form RD estimates would confound program effects with differences in the populations served. We therefore compare the two programs using the covariate-adjusted specifications (column 2 of Tables 13–16), which absorb the observable differences between HV and CC applicants. The difference table is presented in Section 6.5.

As a formal test for manipulation of the running variable, we apply the local-polynomial density test of Cattaneo et al. (2020) at the proposal level; there should be no systematic bunching of proposals just above or just below the cutoff. Figure 2 shows the estimated densities on each side and reports the test result. The test yields $p = 0.075$, failing to reject the null of no discontinuity in density at the conventional 5% level. With only 34 distinct proposal-level scores the test has limited power, but the institutional design also makes strategic score manipulation implausible: the funding score was computed externally by government evaluators, not by the applicants themselves.

Smoothness of the proposal density does not imply smoothness in the number of sampled observations across the score distribution, for two reasons. First, proposals differ in size. Second, we randomly sampled centers stratified by treatment status, program type, urban/rural area, center size, and province to achieve balance across treated and untreated groups. Without stratification, we might have over-sampled the two geographically extreme proposals closest to the threshold, biasing estimates for spatially heterogeneous outcomes.

Figure 2. Test for manipulation in the running variable



Notes: Local-polynomial density estimates on each side of the funding-score cutoff (normalized to zero), with 95% confidence intervals, estimated by `rddensity` (Cattaneo et al., 2020). A robust test for a discontinuity in the density yields $p = 0.075$. The near-uniform spread of proposals on both sides of the threshold is consistent with the external scoring process; strategic manipulation of the score is institutionally implausible. The running variable takes only 34 distinct values, which reduces power of the test.

6 Results

This section first shows how the three inference procedures for the discrete running variable affect the conclusions that can be drawn from the estimates. It then presents results for children's cognitive outcomes, for children's health outcomes, and for mothers' outcomes, and it concludes with a descriptive comparison of the two programs.

6.1 Robustness of inference to the discrete running variable

Table 10 reports our main covariate-adjusted local-linear estimates (identical to column (2) of the main tables) alongside the three inference procedures described in Section 3. The point estimates are unchanged, but the wild cluster bootstrap p-values and the honest confidence intervals are substantially less precise than the clustered analytic standard errors. The wild cluster bootstrap leaves no individual outcome significant at conventional levels. The honest confidence intervals contain zero for every outcome except one: the effect of home visits on the Nelson-Ortiz overall development score, with an interval of $[0.16, 0.68]$. Its wild-cluster-bootstrap p-value is 0.36, so also this effect is not significant under the bootstrap. It is, however, the only effect that meets the robustness standard defined in Section 3.

Table 10. Robustness of inference to the discrete running variable

Outcome	RD est. (LL, $+X$)	p (robust, by cluster)	p (wild bootstrap)	Honest CI (Armstrong-Kolesár)	Local-rand. p
Panel A. Home visits					
Woodcock memory	0.67	0.001	0.149	[-0.50, 1.86]	0.029
Fine motor (pegboard)	0.90	0.000	0.131	[-1.08, 2.46]	0.000
TVIP (vocabulary)	0.63	0.001	0.179	[-1.16, 2.64]	0.119
Nelson-Ortiz total dev.	0.30	0.000	0.358	[0.16, 0.68]	0.002
Anemia	-0.08	0.066	0.194	[-0.44, 0.29]	0.569
Mother stress (CES-D)	-0.69	0.000	0.065	[-2.10, 0.81]	0.000
Mother works	-0.26	0.000	0.231	[-0.35, 0.17]	0.000
Mother hours worked	-8.92	0.000	0.220	[-21.24, 6.18]	0.000
Panel B. Child care					
Woodcock memory	-0.18	0.310	0.557	[-4.00, 5.00]	0.909
Fine motor (pegboard)	2.26	0.000	0.720	[-2.15, 4.60]	0.000
TVIP (vocabulary)	0.32	0.249	0.384	[-9.98, 12.92]	0.000
Nelson-Ortiz total dev.	-0.02	0.728	0.452	[–, –]	0.825
Anemia	0.12	0.027	0.528	[–, –]	1.000
Mother stress (CES-D)	0.12	0.139	0.238	[-1.68, 2.08]	0.753
Mother works	0.10	0.138	0.110	[-3.06, 2.95]	1.000
Mother hours worked	4.82	0.124	0.134	[-102.17, 105.83]	0.397

Notes: Each row is the local effect of the program (relative to no program) at the funding cutoff; continuous outcomes in comparison-group SD units. Column 2 reproduces the main covariate-adjusted local-linear estimate from column (2) of Tables 13–16 (`rdrobust`, MSE-optimal bandwidth, triangular kernel); column 3 its robust bias-corrected p-value, clustered on project $\times$ parish. Column 4 is the wild-cluster-bootstrap p-value for the reduced-form discontinuity over the 34 proposals (Cameron et al., 2008); column 5 the bias-aware honest 95% CI for a discrete running variable at the rule-of-thumb smoothness bound (Kolesár and Rothe, 2018; Armstrong and Kolesár, 2020); column 6 the finite-sample randomization-inference p-value in a ± 5 window with child-level permutation (Cattaneo et al., 2015). As Table 12 shows, the child-level permutation in column 6 overstates the information in the design; proposal-level permutation yields no significant results.

The honest confidence intervals depend on the smoothness bound M, which cannot be chosen from the data. Table 11 therefore reports the intervals at the rule-of-thumb bound and at two and four times that bound, together with the fuzzy bias-aware intervals that account for the estimated first stage (Noack and Rothe, 2024). The interval for the home-visit effect on overall development excludes zero at the rule-of-thumb bound, and continues to do so when the bound is doubled ($[0.03, 0.81]$); zero enters the interval only when the bound is quadrupled. The fuzzy interval, $[0.16, 0.71]$, is nearly identical to the sharp reduced-form interval, as expected with a first stage of 0.97–0.99. For all other home-visit outcomes the intervals contain zero at every reported bound. For child care centers the intervals are wide throughout; for two outcomes the sharp procedure returns no valid interval at the optimal bandwidth, while the fuzzy procedure returns wide intervals that contain zero.

Table 11. Honest confidence intervals: sensitivity to the smoothness bound

	ROT M	CI at ROT M	CI at $2M$	CI at $4M$	Fuzzy CI
Panel A. Home visits					
Woodcock memory	0.222	[-0.50, 1.86]	[-1.38, 2.75]	[-3.15, 4.51]	[-0.50, 1.87]
Fine motor (pegboard)	0.377	[-1.08, 2.46]	[-2.60, 3.98]	[-5.64, 7.02]	[-1.09, 2.47]
TVIP (vocabulary)	0.399	[-1.16, 2.64]	[-2.76, 4.24]	[-5.96, 7.44]	[-1.17, 2.65]
Nelson-Ortiz total dev.	0.033	[0.16, 0.68]	[0.03, 0.81]	[-0.24, 1.08]	[0.16, 0.71]
Anemia	0.100	[-0.44, 0.29]	[-1.05, 0.90]	[-1.87, 1.73]	[-0.44, 0.29]
Mother stress (CES-D)	0.291	[-2.10, 0.81]	[-3.28, 1.99]	[-5.63, 4.34]	[-2.15, 0.84]
Mother works	0.052	[-0.35, 0.17]	[-0.73, 0.40]	[-1.15, 0.82]	[-0.36, 0.17]
Mother hours worked	2.030	[-21.24, 6.18]	[-29.43, 14.37]	[-45.82, 30.75]	[-21.65, 6.31]
Panel B. Child care centers					
Woodcock memory	0.314	[-4.00, 5.00]	[-8.02, 9.02]	[-16.05, 17.06]	[-4.76, 5.77]
Fine motor (pegboard)	0.226	[-2.15, 4.60]	[-5.05, 7.50]	[-10.85, 13.30]	[-4.02, 6.47]
TVIP (vocabulary)	0.861	[-9.98, 12.92]	[-21.00, 23.94]	[-43.04, 45.99]	[-12.22, 15.16]
Nelson-Ortiz total dev.	0.201	–	–	–	[-3.14, 3.65]
Anemia	0.044	–	–	–	[-0.84, 0.90]
Mother stress (CES-D)	0.119	[-1.68, 2.08]	[-3.20, 3.60]	[-6.24, 6.64]	[-2.00, 2.40]
Mother works	0.220	[-3.06, 2.95]	[-5.87, 5.76]	[-11.50, 11.38]	[-3.16, 3.04]
Mother hours worked	7.625	[-102.17, 105.83]	[-199.76, 203.41]	[-394.94, 398.59]	[-105.10, 108.76]

Notes: Bias-aware honest 95% confidence intervals (Kolesár and Rothe, 2018; Armstrong and Kolesár, 2020) for the reduced-form discontinuity, computed with `rdhonest`. Column 2 reports the rule-of-thumb (ROT) smoothness bound M on the second derivative of the conditional expectation function; columns 3–5 the intervals when M is set to one, two and four times the ROT value. The last column reports the fuzzy bias-aware interval, which accounts for the estimated first stage (Noack and Rothe, 2024). A dash indicates that the procedure returns no valid interval (zero effective observations at the optimal bandwidth). Continuous outcomes in comparison-group SD units.

Local-randomization inference requires that proposals within the chosen window are as good as randomly assigned to the two sides of the cutoff, and the appropriate permutation unit is the proposal. Table 12 reports randomization p-values for windows from ± 5 to ± 10, permuting at the child level and at the proposal level, together with the number of proposals in each window. The window composition explains the results. The ± 5 window contains three proposals in the home-visit sample (one below the cutoff, two above) and two in the child-care sample, so a proposal-level permutation test cannot produce a p-value below one third; windows of ± 3 and ± 4 contain the same proposals. At ± 7 and ± 10 the proposal-level p-values range from 0.13 to 1, with none significant. The small child-level p-values in Table 10 thus reflect the assumption that children within a proposal provide independent variation, which they do not. We conclude that local-randomization inference is uninformative in this design, and we do not count it as support for any finding. We note that no covariate-balanced window exists: the funding score is correlated with the community wealth index by construction, so the as-if-random assumption is questionable even apart from the small number of proposals.

Table 12. Local-randomization inference: window sensitivity and permutation level

Outcome	Child-level p ±5	±7	±10	Proposal-level p ±7	±10	Proposals in ±10
Panel A. Home visits						
Woodcock memory	0.029	0.000	0.000	0.657	0.380	4/2
Fine motor (pegboard)	0.000	0.000	0.000	0.330	0.133	4/2
TVIP (vocabulary)	0.119	0.000	0.007	0.657	0.754	4/2
Nelson-Ortiz total dev.	0.002	0.006	0.016	0.681	0.652	4/2
Anemia	0.569	0.014	0.001	0.330	0.133	4/2
Mother stress (CES-D)	0.000	0.000	0.000	0.330	0.396	4/2
Mother works	0.000	0.000	0.002	0.330	0.537	4/2
Mother hours worked	0.000	0.000	0.001	0.330	0.266	4/2
Panel B. Child care centers						
Woodcock memory	0.909	0.909	0.175	–	0.413	2/3
Fine motor (pegboard)	0.000	0.000	0.681	–	1.000	2/3
TVIP (vocabulary)	0.000	0.000	0.207	–	0.814	2/3
Nelson-Ortiz total dev.	0.825	0.825	0.938	–	0.814	2/3
Anemia	1.000	1.000	0.139	–	0.379	2/3
Mother stress (CES-D)	0.753	0.753	0.004	–	0.379	2/3
Mother works	1.000	1.000	0.000	–	0.410	2/3
Mother hours worked	0.397	0.397	0.000	–	0.197	2/3

Notes: Finite-sample randomization-inference p-values (Cattaneo et al., 2015) for the indicated windows around the cutoff (in normalized score units), computed with `rdrandinf` (2,000 permutations). Child-level columns permute assignment across children; proposal-level columns collapse outcomes to proposal means within the window and permute across proposals, the level at which treatment is assigned. A dash indicates that fewer than four proposals fall in the window, in which case the permutation distribution is too coarse to be informative (with three proposals the smallest attainable p-value is one third). Windows of ±3 and ±4 contain the same proposals as ±5. The last column reports the number of proposals below/above the cutoff in the ±10 window.

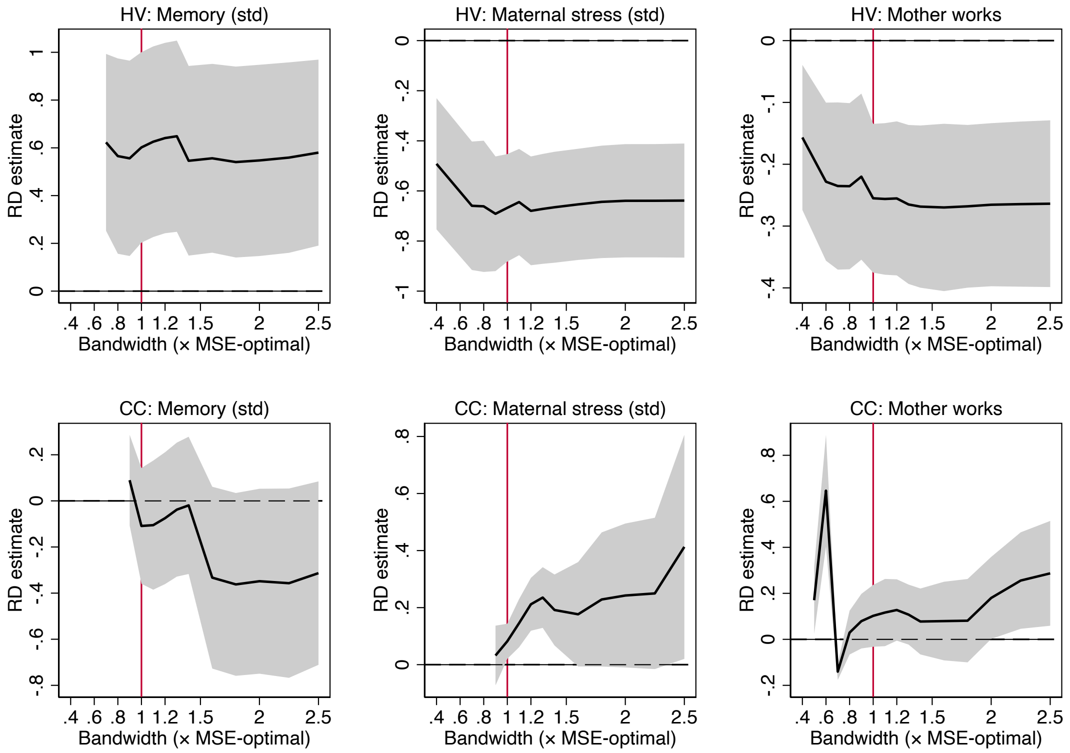
Figure 3 examines sensitivity to the choice of bandwidth. Each panel traces the bias-corrected local-linear estimate and its 95% robust confidence interval as the bandwidth varies from 40% to 250% of the MSE-optimal value (marked by the vertical red line). The estimates are stable across the full range for the home-visit outcomes (memory, maternal stress, and maternal employment). For child care centers the estimates are noisier, especially at the smallest bandwidths, reflecting the much smaller sample; but they show no systematic trend that would overturn the estimates at the MSE-optimal bandwidth.

We conclude from this section that one effect meets our robustness standard: home visits raise children's overall development. The home-visit effects on memory, fine motor skills, maternal stress and maternal labor supply are significant under conventional clustered inference but not under the wild cluster bootstrap or the honest confidence intervals; we treat them as suggestive throughout. No child-care effect meets either standard. The following subsections present the estimates and apply these labels.

6.2 *Children's cognitive and motor outcomes*

Figure 4 plots binned sample means of three standardized cognitive and motor outcomes against the normalized proposal funding score, separately for home visits (left column)

Figure 3. Bandwidth sensitivity of main estimates



Notes: Each panel plots the bias-corrected RD estimate (solid line) with 95% robust confidence interval (shaded band) as the bandwidth varies from 0.4× to 2.5× the MSE-optimal value. The vertical red line marks the MSE-optimal bandwidth; the dashed horizontal line marks zero. Estimates use local linear with covariate adjustment (`rdrobust`, triangular kernel, clustered on project × parish). Rows: home visits (top) and child care centers (bottom). Columns: memory test (left), maternal stress (center), mother works (right).

and child care centers (right column), estimated with `rdplot` (Calonico et al., 2014). Each bin represents the mean outcome of children whose proposal score falls within an evenly-spaced interval; the curves are local-linear fits on each side of the cutoff.

The slopes within each side of the cutoff are shallow and mostly negative, reflecting that the proposal score rewards poorer communities, so children from higher-scoring proposals tend to have lower baseline outcomes. For home visits the local-linear fits display a clear upward jump at the cutoff for all three outcomes. For child care centers the discontinuity is smaller and less consistent across outcomes.

We further investigate this in Tables 13 and 14, which report local-linear RD estimates of the impact of home visits and child care centers on all cognitive, motor and health outcomes. The top panel of each table groups the tests administered to children older than 36 months, the second panel the Nelson-Ortiz dimensions for children younger than 60 months, and the bottom panel the health outcomes. Table 13 reports estimates for home visits and Table 14 for child care centers, based on equations (1) and (2), respectively. Each table reports three specifications: local linear with the MSE-optimal bandwidth and no covariates (column 1), the same with covariate adjustment (column 2), and local linear with the CER-optimal bandwidth and covariates (column 3).[10]

The results for home visits in the top panel reveal the same pattern as Figure 4. Home visits have a positive point estimate on most cognitive and motor outcomes in both specifications, and adding covariates produces very similar but more precise estimates. Focusing on the specification with covariates, all three tests for children older than 36 months, together with fine motor skills and the overall development index for children under 60 months, are significantly positive at conventional levels. The point estimates are sizeable: the standardized cognitive and motor tests improve by between 0.6 and 0.9 of a standard deviation, and the overall Nelson-Ortiz index, measured as the probability of scoring above the age-group mean, rises by about 30 percentage points relative to a control-group base of 17 percent. These conventional-inference results should be read against Section 6.1: only the effect on the overall development index has an honest confidence interval that excludes zero, and it does so also within the age range that is balanced across the cutoff (Table 8). The individual test results are not robust to the wild cluster bootstrap or the honest confidence intervals, and we treat them as suggestive.

For child care centers the estimates are mixed in sign and, with the exception of fine motor skills for children over 36 months, mostly insignificant. That one large positive estimate is not credible as a stable effect: it is highly sensitive to the bandwidth (it more than halves when the bandwidth is doubled) and does not survive the wild cluster bootstrap or the honest confidence interval (Table 10). Across the cognitive and motor outcomes there is thus no consistent or robust pattern, and we find no reliable evidence that child care centers improve children's cognitive development.[11]

6.3 Children's health outcomes

We next look at the impact of the programs on children's health outcomes; the estimates are in the bottom panels of Tables 13 and 14. For home visits, the covariate-adjusted estimates point to a reduction in the incidence of anemia of about 8 percentage points, relative to a control-group rate of 0.46 (a reduction of roughly one-sixth). The point estimates for underweight and below height for age are positive (that is, in the adverse direction) and

[10]The data also contain a variable indicating households' poverty. This variable cannot be included as a control variable because it has been measured after the intervention and might therefore have been affected by it.

[11]We also inquired whether impacts are different for boys than for girls, and for children above and below 24 months old. Neither for home visits nor for child care centers, do we find any evidence of heterogeneity of impacts across these groups.

Table 13. Effects of home visits on cognitive, motor and health outcomes

	(1) LL, MSE	(2) LL, MSE+X	(3) LL, CER+X
A. Cognitive (children >36 months)			
Memory	0.682***	0.673***	0.688***
	(0.191)	(0.194)	(0.189)
Language	0.745**	0.631***	0.588***
	(0.354)	(0.189)	(0.168)
Fine motor	0.838***	0.898***	0.929***
	(0.152)	(0.136)	(0.125)
B. Development (children ≤60 months)			
Gross motor	0.151	0.189	0.164
	(0.177)	(0.146)	(0.146)
Fine motor	0.238***	0.116***	0.164***
	(0.024)	(0.028)	(0.026)
Language	0.043	-0.024	0.035
	(0.079)	(0.042)	(0.032)
Social/personal	0.231	0.222*	0.218*
	(0.199)	(0.133)	(0.127)
Total	0.436***	0.300***	0.400***
	(0.088)	(0.086)	(0.077)
C. Health			
Anemia	-0.096	-0.078*	-0.101***
	(0.076)	(0.042)	(0.036)
Underweight (WAZ<-2)	0.072**	0.065**	0.065**
	(0.029)	(0.027)	(0.027)
Below height (HAZ<-2)	0.130	0.140**	0.146***
	(0.108)	(0.057)	(0.055)

Notes: Each cell reports the bias-corrected RD point estimate (top) with the robust standard error in parentheses (bottom), obtained with `rdrobust` (Calonico et al., 2014). (1) local linear (p=1), MSE-optimal bandwidth, no covariates; (2) local linear, MSE-optimal, with covariate adjustment (Calonico et al., 2019); (3) local linear, CER-optimal bandwidth, with covariates. Local quadratic specifications are not feasible because bandwidth selection fails with the discrete running variable (34 mass points); see Kolesár and Rothe (2018). All specifications use a triangular kernel and standard errors clustered on project × parish (98 clusters). Covariates: child sex, number of children under five, urban, welfare benefit, mother's age quadratic, mother's and household head's schooling, mother's vocabulary score, two parenting-practice dummies, ethnicity, household composition, and mother's literacy. Standardized outcomes are in standard-deviation units. Bandwidths (h) and effective sample sizes (N) vary by outcome and specification. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 14. Effects of child care centers on cognitive, motor and health outcomes

	(1) LL, MSE	(2) LL, MSE+X	(3) LL, CER+X
A. Cognitive (children >36 months)			
Memory	0.577***	-0.176	0.717***
	(0.210)	(0.173)	(0.093)
Language	1.531***	0.321	1.568***
	(0.403)	(0.278)	(0.224)
Fine motor	1.352***	2.257***	2.266***
	(0.135)	(0.159)	(0.151)
B. Development (children ≤60 months)			
Gross motor	-0.163***	-0.074**	-0.011
	(0.027)	(0.035)	(0.039)
Fine motor	0.179*	-0.003	-0.094**
	(0.104)	(0.046)	(0.039)
Language	0.151**	0.061	0.103*
	(0.068)	(0.062)	(0.062)
Social/personal	0.223*	0.073	0.122
	(0.133)	(0.061)	(0.080)
Total	0.283***	-0.019	0.169***
	(0.088)	(0.054)	(0.049)
C. Health			
Anemia	0.068	0.124**	0.110**
	(0.078)	(0.056)	(0.046)
Underweight (WAZ<-2)	0.225***	0.262***	0.290***
	(0.043)	(0.029)	(0.026)
Below height (HAZ<-2)	-0.071	0.384***	0.384***
	(0.159)	(0.070)	(0.092)

Notes: Each cell reports the bias-corrected RD point estimate (top) with the robust standard error in parentheses (bottom), obtained with `rdrobust` (Calonico et al., 2014). (1) local linear (p=1), MSE-optimal bandwidth, no covariates; (2) local linear, MSE-optimal, with covariate adjustment (Calonico et al., 2019); (3) local linear, CER-optimal bandwidth, with covariates. Local quadratic specifications are not feasible because bandwidth selection fails with the discrete running variable (34 mass points); see Kolesár and Rothe (2018). All specifications use a triangular kernel and standard errors clustered on project × parish (98 clusters). Covariates: child sex, number of children under five, urban, welfare benefit, mother's age quadratic, mother's and household head's schooling, mother's vocabulary score, two parenting-practice dummies, ethnicity, household composition, and mother's literacy. Standardized outcomes are in standard-deviation units. Bandwidths (h) and effective sample sizes (N) vary by outcome and specification. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

significant under the clustered analytic standard errors. None of these health effects is robust, however. Once we apply the inference methods for the discrete running variable, the anemia effect loses significance (Table 10), and the same holds for the underweight and below-height estimates: their wild-cluster-bootstrap p-values are 0.14 and 0.40 and their honest confidence intervals comfortably include zero. We therefore do not interpret these as evidence that home visits affect children's physical health in either direction. For child care centers the point estimates also indicate higher rates of underweight and below height, but these likewise do not survive inference designed for the discrete running variable (wild-bootstrap p-values of 0.48 and 0.24, honest confidence intervals spanning zero). Overall, we find no robust effect of either program on children's measured health.

6.4 Mothers' outcomes

Early childhood interventions potentially also have an impact on the way mothers interact with their children and on mothers' psychological well-being and labor market outcomes. Tables 15 and 16 present results for maternal wellbeing and labor market outcomes.

The parenting and wellbeing results for home visits are in the top panel of Table 15. Focusing on the covariate-adjusted specification, the point estimate indicates that home visits make mothers more responsive to their children (a reduction of about half a standard deviation in the non-responsiveness index), though this effect is significant only in the specification without covariates and, at the ten-percent level, under the CER bandwidth. Home visits have no significant effect on whether the mother reads to her child or on the presence of learning materials (books, clay, toys) at home. The clearest maternal result is for psychological wellbeing: home visits reduce stress and depression by about 0.7 of a standard deviation, an effect that is significant across all three specifications. For child care centers (Table 16) the corresponding point estimates for responsiveness and for maternal stress are close to zero and not significant in the covariate-adjusted specification, so the favourable wellbeing pattern is specific to home visits. As with the child outcomes, the home-visit effect on maternal stress is significant under conventional clustered inference but not under the wild cluster bootstrap or the honest confidence interval (Table 10), and we treat it as suggestive.

The descriptive comparison of the two programs, reported in Section 6.5 (Table 17), points in the same direction: relative to child care centers, home visits are associated with more responsive parenting and lower maternal stress.

The maternal labor-market results appear in the bottom panels of Tables 15 and 16. The covariate-adjusted point estimates have the expected opposite signs. Child care centers raise mothers' labor supply (about 10 percentage points more likely to work and roughly 5 additional hours per month), while home visits lower it, by about 26 percent-

Table 15. Effects of home visits on parenting and maternal labor market outcomes

	(1) LL, MSE	(2) LL, MSE+X	(3) LL, CER+X
A. Parenting			
Non-responsive (HOME, std.)	-0.625**	-0.502	-0.531*
	(0.319)	(0.310)	(0.291)
Mother reads to child	-0.031	-0.009	-0.003
	(0.093)	(0.039)	(0.042)
Learning materials at home	-0.122*	-0.076	-0.103
	(0.063)	(0.057)	(0.077)
B. Maternal wellbeing and labor market			
Stress/depression (CES-D, std.)	-0.742***	-0.689***	-0.763***
	(0.143)	(0.116)	(0.130)
Mother works	-0.285***	-0.257***	-0.250***
	(0.107)	(0.061)	(0.059)
Hours worked per month	-10.788***	-8.918***	-8.608***
	(3.398)	(1.817)	(1.712)
Mother's earnings	-11.600	-37.278***	-38.504***
	(15.336)	(8.897)	(8.255)
Household head's earnings	24.723	-6.705	-9.947
	(30.275)	(15.213)	(14.784)

Notes: Each cell reports the bias-corrected RD point estimate (top) with the robust standard error in parentheses (bottom), obtained with `rdrobust` (Calonico et al., 2014). (1) local linear (p=1), MSE-optimal bandwidth, no covariates; (2) local linear, MSE-optimal, with covariate adjustment (Calonico et al., 2019); (3) local linear, CER-optimal bandwidth, with covariates. Local quadratic specifications are not feasible because bandwidth selection fails with the discrete running variable (34 mass points); see Kolesár and Rothe (2018). All specifications use a triangular kernel and standard errors clustered on project × parish (98 clusters). Covariates: child sex, number of children under five, urban, welfare benefit, mother's age quadratic, mother's and household head's schooling, mother's vocabulary score, two parenting-practice dummies, ethnicity, household composition, and mother's literacy. Standardized outcomes are in standard-deviation units. Bandwidths (h) and effective sample sizes (N) vary by outcome and specification. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 16. Effects of child care centers on parenting and maternal labor market outcomes

	(1) LL, MSE	(2) LL, MSE+X	(3) LL, CER+X
A. Parenting			
Non-responsive (HOME, std.)	-0.232**	0.062	-0.051
	(0.093)	(0.080)	(0.082)
Mother reads to child	-0.174***	-0.053***	-0.061***
	(0.031)	(0.016)	(0.014)
Learning materials at home	0.415***	0.602***	0.572***
	(0.072)	(0.066)	(0.055)
B. Maternal wellbeing and labor market			
Stress/depression (CES-D, std.)	0.228	0.120	0.050
	(0.144)	(0.081)	(0.040)
Mother works	-0.031	0.102	0.081
	(0.081)	(0.069)	(0.074)
Hours worked per month	5.000*	4.823	5.982
	(2.782)	(3.138)	(3.663)
Mother's earnings	30.330***	10.652	12.858*
	(3.963)	(6.891)	(7.636)
Household head's earnings	168.041***	108.277***	114.855***
	(6.446)	(7.736)	(7.072)

Notes: Each cell reports the bias-corrected RD point estimate (top) with the robust standard error in parentheses (bottom), obtained with `rdrobust` (Calonico et al., 2014). (1) local linear (p=1), MSE-optimal bandwidth, no covariates; (2) local linear, MSE-optimal, with covariate adjustment (Calonico et al., 2019); (3) local linear, CER-optimal bandwidth, with covariates. Local quadratic specifications are not feasible because bandwidth selection fails with the discrete running variable (34 mass points); see Kolesár and Rothe (2018). All specifications use a triangular kernel and standard errors clustered on project × parish (98 clusters). Covariates: child sex, number of children under five, urban, welfare benefit, mother's age quadratic, mother's and household head's schooling, mother's vocabulary score, two parenting-practice dummies, ethnicity, household composition, and mother's literacy. Standardized outcomes are in standard-deviation units. Bandwidths (h) and effective sample sizes (N) vary by outcome and specification. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

age points and roughly 9 hours per month. For home visits these labor-supply estimates are significant under the clustered analytic standard errors; for child care centers they are not even conventionally significant in the local-linear specification. On earnings, home visits are associated with a significant reduction in the mother's own earnings and an insignificant effect on the household head's earnings, whereas child care centers show a positive and significant effect on the household head's earnings but an insignificant effect on the mother's own earnings.[12] These labor-market effects must again be read against the discrete-running-variable inference (Table 10): none survives the wild cluster bootstrap, and the home-visit effects on labor supply, while significant under conventional clustered inference, we treat as suggestive. The descriptive program comparison is reported in Section 6.5.

6.5 Comparing the two programs: descriptive evidence

Table 17 places the two sets of estimates side by side. For each outcome we report the HV$-$CC difference in the covariate-adjusted RD estimates (the underlying estimates are in column (2) of Tables 13–16) and the p-value for the null of equal effects. As explained in Section 3, we treat these differences as descriptive. Covariate adjustment absorbs the observable differences between HV and CC applicants documented in Table 9, but it does not account for unobservables, and no adjustment can separate the effect of program type from the effects of intensity, delivery and population.

[12]The larger effect on the household head's earnings than on the mother's own earnings most likely reflects that many poor households in Ecuador earn their income in the informal sector through activities in which household members work together.

Table 17. Home visits minus child care centers: covariate-adjusted RD estimates

	HV−CC	p(diff= 0)
A. Cognitive (children >36 months)		
Memory (std.)	0.849***	0.001
	(0.260)	
Language (std.)	0.311	0.356
	(0.336)	
Fine motor (std.)	-1.359***	0.000
	(0.209)	
B. Development (children ≤60 months)		
Gross motor (std.)	0.263*	0.079
	(0.150)	
Fine motor (std.)	0.119**	0.027
	(0.054)	
Language (std.)	-0.085	0.254
	(0.075)	
Social/personal (std.)	0.150	0.307
	(0.146)	
Total (std.)	0.319***	0.002
	(0.101)	
C. Health		
Anemia	-0.202***	0.004
	(0.070)	
Underweight (WAZ<−2)	-0.197***	0.000
	(0.039)	
Below height (HAZ<−2)	-0.245***	0.007
	(0.090)	
D. Parenting		
Non-responsive (HOME, std.)	-0.564*	0.078
	(0.321)	
Mother reads to child	0.045	0.292
	(0.042)	
Learning materials at home	-0.678***	0.000
	(0.088)	
E. Maternal wellbeing and labor market		
Stress/depression (CES-D, std.)	-0.809***	0.000
	(0.141)	
Mother works	-0.360***	0.000
	(0.092)	
Hours worked per month	-13.741***	0.000
	(3.626)	
Mother's earnings	-47.930***	0.000
	(11.254)	
Household head's earnings	-114.981***	0.000
	(17.067)	

Notes: Difference in covariate-adjusted RD estimates (HV minus CC), specification (2) of Tables 13–16. Standard error of the difference in parentheses, assuming independence of the two samples. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

The estimates point to a broadly consistent pattern. On children's development, the

HV$-$CC difference is positive and significant for Woodcock memory and for the overall Nelson-Ortiz development index, favouring home visits. For the pegboard fine-motor test the difference is large and negative, but this is driven by the implausibly large and non-robust child-care fine-motor estimate discussed above rather than by a home-visit shortfall; the difference in language is not significant. On health, the differences favour home visits (lower anemia, underweight and below height). For maternal outcomes the pattern reverses: child care centers are associated with greater maternal labor supply and earnings, while home visits are associated with lower maternal stress and labor supply, with the differences in stress, labor supply and earnings all sizeable.

A natural concern is that the opposite-signed labor-supply estimates reflect the targeting of the two programs rather than their effects: the guidelines describe child care centers as targeted to communities where mothers work and home visits to communities where mothers stay home. We can examine the pre-existing differences directly by comparing mothers in the unfunded (below-cutoff) communities of the two programs, whose outcomes are unaffected by treatment. Table 18 reports this comparison. It shows the opposite of the targeting described in the guidelines: within ± 15 score units of the cutoff, 47 percent of mothers in unfunded home-visit communities work, against 21 percent in unfunded child-care communities, and the same ordering holds for hours and earnings. The communities served by the two programs thus differ substantially in baseline maternal employment, which reinforces the descriptive reading of Table 17. At the same time, the direction of the difference works against the estimated pattern: home visits reduce maternal labor supply from a high baseline, and child care centers raise it from a low one. Pre-existing differences of this sign cannot mechanically generate the estimated effects. In addition, adding community-level covariates (population, poverty, access to water and sanitation, schooling) to the RD specifications leaves the home-visit labor-supply effects intact and, if anything, strengthens the child-care labor-supply estimates (Appendix Table B3).

Table 18. Mothers' labor-market outcomes in unfunded communities, by program type

	All unfunded			Within ±15 of cutoff		
	HV	CC	Diff.	HV	CC	Diff.
Mother works	0.43	0.33	0.10	0.47	0.21	0.26***
			(0.06)			(0.04)
Mother hours worked	13.45	9.50	3.95*	14.57	5.52	9.04***
			(2.11)			(1.60)
Mother stress (CES-D)	0.12	-0.08	0.19*	0.03	-0.09	0.12
			(0.11)			(0.16)
Mother's earnings	71.69	43.90	27.79***	76.38	36.55	39.83***
			(7.06)			(9.68)

Notes: Mean outcomes of mothers in communities whose proposals fell below the funding threshold (no program was implemented), separately for home-visit (HV) and child-care (CC) proposals. These mothers are untreated, so differences between the two columns reflect pre-existing differences between the communities the two programs applied for, not program effects. Standard errors of the differences (in parentheses) clustered on project × parish. Stress in standard-deviation units; earnings in US dollars per month. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Taken together, and bearing in mind that these differences are descriptive and inherit the fragility documented in Section 6.1, the pattern of point estimates is consistent with a trade-off: home visits invest in child development while their communities show lower maternal labor supply, whereas child care centers are associated with more maternal work and weaker and less consistent child-development gains. Our design cannot establish this trade-off as causal at the cross-program level.

7 Conclusions

This paper evaluates two national early childhood programs in Ecuador, home visits and child care centers, funded through the same proposal-scoring mechanism. Each program is evaluated against no intervention through a regression discontinuity design, on a broad range of outcomes: children's cognitive and motor development, children's health, parenting styles, mothers' labor supply and income and mothers' psychological well-being. The evidence base on early childhood programs in developing countries is by now large. Our setting adds a parallel evaluation of a home-visit and a center-based program at national scale, with the same forcing variable, the same outcome measures and the same estimation method, together with a transparent treatment of inference when the forcing variable takes only 34 proposal-level values.

The main finding is that home visits raise children's overall development. The estimated effect is an increase of 30 percentage points in the probability of scoring above the age-norm mean of the Nelson-Ortiz scale, relative to a control-group base of 17 percent.

Its bias-aware honest confidence interval excludes zero, also when the smoothness bound is doubled, when the fuzzy correction is applied, and when the sample is restricted to an age range that is balanced across the cutoff. The wild cluster bootstrap over the 34 proposals does not reject at conventional levels, so we do not regard this finding as conclusive. The effects of home visits on memory, fine motor skills, maternal stress and maternal labor supply are significant under conventional clustered inference only, and we treat them as suggestive. The effects of child care centers on children vary in sign and are not robust, and the apparent effects of child care centers on mothers' labor supply and stress are not significant under the local-linear specification. We also find that local-randomization inference, when the permutation respects the proposal level at which treatment is assigned, is uninformative in this design; child-level permutation overstates the information content of the data. That individual cognitive-test effects are fragile is consistent with the scale-up literature: Andrew et al. (2018) find that gains from a scalable home-stimulation program in Colombia fade out within two years, and Carneiro et al. (2024) find positive but modest effects of a national parenting program in Chile. The cost of a child care center per child is also almost five times as high as the cost of a home visit.

The results are consistent with the framework of Section 2.3. The most robust effect, on overall child development, arises in the program that works directly on the quality of mother-child interaction. The suggestive effects also follow the predicted margins: lower stress and lower labor supply for home-visit mothers, and more maternal work in child-care communities. We stress that the child-care side of this interpretation rests on estimates that do not survive our most demanding inference procedures.

This discussion makes explicit that our findings are conditional on the quality of the early childhood interventions included in our design. We emphasize that our analysis looks at the effects of programs that were above the minimum quality standards required to receive funding. Moreover, the home visit programs and child care centers that we evaluate are of the same quality level (measured by the scores that the proposals received in the contest for quality indicators). At the same time, lower quality of child care centers in Ecuador than those operated elsewhere might explain why we do not find the positive effects of child care centers on children's development that studies for Norway and Germany report (Havnes and Mogstad, 2011; Felfe and Lalive, 2012). The absence of clear child-development gains from child care centers is, however, in line with the results reported for Canada (Baker et al., 2008).

The pattern of point estimates is consistent with a trade-off between child development and maternal labor supply: home visits benefit children while their communities show lower maternal labor supply, whereas child care centers are associated with greater maternal labor-market participation and income. We present this pattern as descriptive.

The two programs serve different populations, the cross-program differences rest on a selection-on-observables assumption, and most of the individual components do not survive the inference procedures for the discrete running variable. Yet the possibility of such a trade-off is relevant for policy makers and funding agencies, since children and women in poor families are both vulnerable groups, and it deserves examination in designs that can identify it, for instance experiments that randomize program type across comparable communities.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude (Anthropic) to assist with editing the text, developing the replication code, and checking the consistency of code, tables, and text. The authors reviewed and verified all output and take full responsibility for the content.

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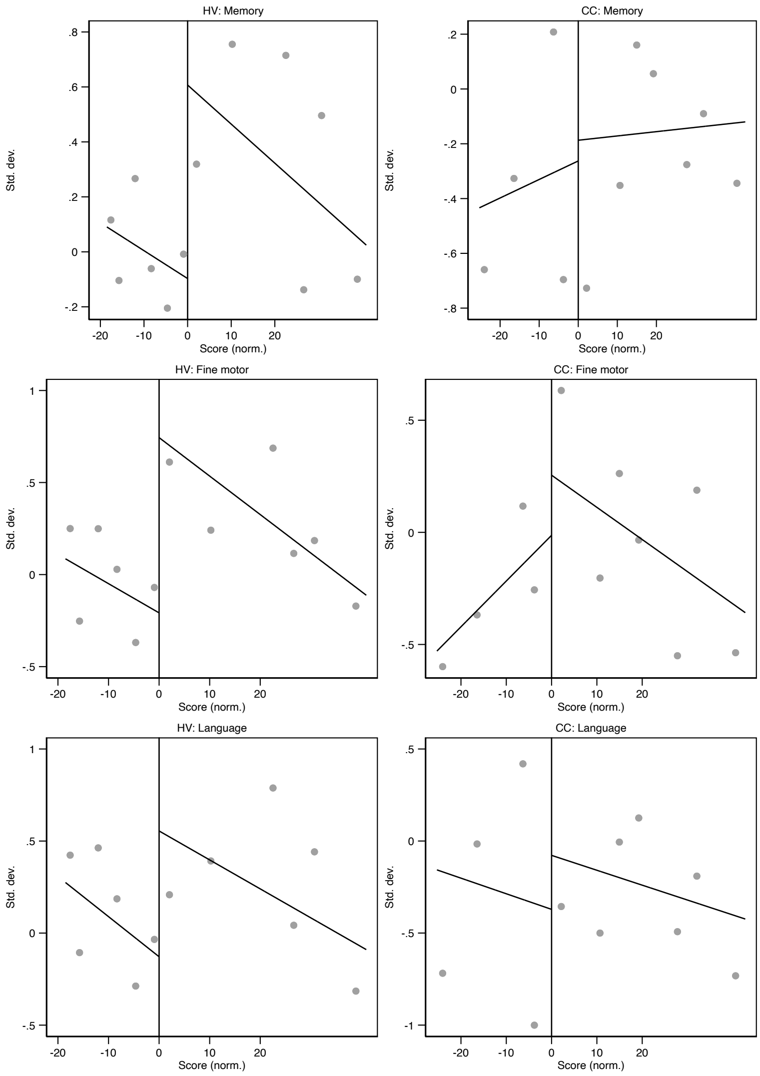
Figure 4. RD plots: cognitive and motor outcomes by type of intervention



Notes: Binned sample means (circles) and local-linear polynomial fits on each side of the cutoff (solid lines), estimated with `rdplot` (Calonico et al., 2014). Running variable normalized to zero at the funding cutoff. Left column: home visits (CNH). Right column: child care centers (CDI/FODI). All outcomes standardized to mean zero, unit standard deviation (children >36 months): Woodcock-Johnson memory test (top), pegboard fine-motor test (middle), TVIP vocabulary test (bottom). Displayed window: ± 20 normalized score units around the cutoff; local-linear fit uses the full data range.

A Outcome definitions and measurement

This appendix describes the outcome measures in more detail. All tests and measurements were administered by trained data-collection teams during a single home visit between September and December 2008, using the same protocol on both sides of the funding threshold.

Tests for children older than 36 months. Three instruments were administered to children older than 36 months. The TVIP (Test de Vocabulario en Imágenes Peabody), the Spanish version of the Peabody Picture Vocabulary Test, measures receptive vocabulary. The Woodcock-Muñoz test measures long-term memory. The Pegboard test measures fine motor skills. All three produce age-normed scores, which we standardize to mean zero and standard deviation one in the sample. These instruments are validated for children older than 36 months, so their samples are restricted by design, not by non-response; completion among eligible children is 96 to 97 percent.

The Nelson-Ortiz scale. For children up to 60 months we use the Escala Abreviada de Desarrollo (Ortiz, 1999), an interviewer-administered screening instrument widely used in Latin America. The scale contains items in four domains: gross motor development, fine motor (adaptive) development, hearing and language, and personal-social development. Each child is assessed on the items for the child's age band, and the domain score is the number of items passed. The total score is the sum of the four domain scores. The scale's published norm tables, defined for ten age bands between 1 and 72 months, classify the total score into four categories: alert, medium, medium-high and high, where the boundary between medium and medium-high corresponds to the mean of the norm population in the child's age band. Our binary outcome, overall development, equals one if the child is classified medium-high or high, that is, above the norm-population mean for the child's age band. The four domain outcomes are constructed analogously from the domain-specific norm tables. Because the reference is the scale's external norm population, the outcome definition does not depend on the age composition of our sample. The test was completed for 98 percent of sampled children; children younger than 1 month are outside the scale's range. We restrict the analysis to children up to 60 months because the two oldest norm bands share the same cutoffs.

Health. Height and weight were measured during the visit and converted to height-for-age and weight-for-age z-scores; below height and underweight are indicators for z-scores below -2. Anemia is measured from a blood sample, with hemoglobin adjusted for altitude. The blood sample was obtained for 86 percent of the children, the largest source of item non-response in the data.

Mothers. Psychological wellbeing is measured with the Center for Epidemiological Studies Depression scale (CES-D), standardized in the sample. Responsiveness is measured with the HOME scale, which is based on the interviewer's observation of the mother's interaction with the child during the visit; we cannot rule out that interviewer judgment affects this measure, but the protocol was identical on both sides of the threshold. Labor-market outcomes (participation, hours per month, earnings) are self-reported in the interview.

B Additional tables

Table B1. Effective support near the cutoff, by outcome

Outcome	Bandwidth	Proposals in bw below	Proposals in bw above	Children in bw below	Children in bw above
Panel A. Home visits					
Woodcock memory	17.3	6	3	614	154
TVIP (vocabulary)	14.9	5	3	435	156
Fine motor (pegboard)	22.3	7	3	732	155
Gross motor (NO)	16.4	6	3	399	143
Fine motor (NO)	15.6	5	3	279	145
Language (NO)	13.9	5	3	277	145
Social/personal (NO)	12.9	5	3	279	144
Nelson-Ortiz total dev.	13.8	5	3	280	145
Anemia	12.2	4	3	234	185
Underweight	19.4	7	3	812	196
Below height	13.2	5	3	475	192
Non-responsive (HOME)	22.9	7	4	820	316
Mother reads to child	21.2	7	3	830	199
Learning materials	13.5	5	3	493	199
Mother stress (CES-D)	19.6	7	3	829	199
Mother works	23.0	7	4	830	317
Mother hours worked	23.5	7	4	830	316
Mother's earnings	16.6	6	3	706	199
Head's earnings	21.3	7	3	829	199
Panel B. Child care centers					
Woodcock memory	17.6	4	4	300	162
TVIP (vocabulary)	16.2	3	4	200	162
Fine motor (pegboard)	14.4	2	3	160	104
Gross motor (NO)	15.3	3	4	103	139
Fine motor (NO)	16.2	3	4	100	140
Language (NO)	18.9	4	4	159	138
Social/personal (NO)	19.3	4	5	161	180
Nelson-Ortiz total dev.	18.4	4	4	161	140
Anemia	18.5	4	4	284	172
Underweight	21.2	4	6	333	301
Below height	17.2	3	4	217	190
Non-responsive (HOME)	15.2	2	4	178	184
Mother reads to child	14.5	2	3	178	127
Learning materials	19.3	4	5	341	241
Mother stress (CES-D)	18.5	4	4	341	195
Mother works	22.3	4	6	341	305
Mother hours worked	21.1	4	6	341	304
Mother's earnings	21.0	4	6	341	305
Head's earnings	17.6	4	4	340	195

Notes: For each outcome, the MSE-optimal bandwidth of the column (2) specification of Tables 13–16 (in normalized score units), the number of proposals whose score falls within the bandwidth on each side of the cutoff, and the effective number of children. Because the score is constant within a proposal, the number of distinct score values equals the number of proposals. The score mass points nearest to the cutoff lie at -0.7 and $+0.1$ normalized units in the home-visit sample and at -2.7 and $+0.7$ in the child-care sample.

Table B2. Clustering at the assignment level

Outcome	RD est.	Cluster: project×parish (98) (s.e.)	p	Cluster: proposal (34) (s.e.)	p
Panel A. Home visits					
Woodcock memory	0.673	(0.194)	0.001	(0.185)	0.000
TVIP (vocabulary)	0.631	(0.189)	0.001	(0.132)	0.000
Fine motor (pegboard)	0.898	(0.136)	0.000	(0.183)	0.000
Gross motor (NO)	0.189	(0.146)	0.194	(0.041)	0.000
Fine motor (NO)	0.116	(0.028)	0.000	(0.042)	0.002
Language (NO)	-0.024	(0.042)	0.560	(0.029)	0.951
Social/personal (NO)	0.222	(0.133)	0.095	(0.031)	0.000
Nelson-Ortiz total dev.	0.300	(0.086)	0.000	(0.051)	0.000
Anemia	-0.078	(0.042)	0.066	(0.045)	0.006
Underweight	0.065	(0.027)	0.017	(0.022)	0.003
Below height	0.140	(0.057)	0.014	(0.053)	0.014
Non-responsive (HOME)	-0.502	(0.310)	0.106	(0.398)	0.106
Mother reads to child	-0.009	(0.039)	0.828	(0.039)	0.734
Learning materials	-0.076	(0.057)	0.188	(0.074)	0.296
Mother stress (CES-D)	-0.689	(0.116)	0.000	(0.075)	0.000
Mother works	-0.257	(0.061)	0.000	(0.048)	0.000
Mother hours worked	-8.918	(1.817)	0.000	(0.962)	0.000
Mother's earnings	-37.278	(8.897)	0.000	(4.505)	0.000
Head's earnings	-6.705	(15.213)	0.659	(7.477)	0.445
Panel B. Child care centers					
Woodcock memory	-0.176	(0.173)	0.310	(0.110)	0.066
TVIP (vocabulary)	0.321	(0.278)	0.249	(0.185)	0.000
Fine motor (pegboard)	2.257	(0.159)	0.000	(0.165)	0.000
Gross motor (NO)	-0.074	(0.035)	0.034	(0.026)	0.000
Fine motor (NO)	-0.003	(0.046)	0.949	(0.010)	0.905
Language (NO)	0.061	(0.062)	0.327	(0.044)	0.188
Social/personal (NO)	0.073	(0.061)	0.234	(0.061)	0.082
Nelson-Ortiz total dev.	-0.019	(0.054)	0.728	(0.026)	0.113
Anemia	0.124	(0.056)	0.027	(0.021)	0.000
Underweight	0.262	(0.029)	0.000	(0.011)	0.000
Below height	0.384	(0.070)	0.000	(0.090)	0.000
Non-responsive (HOME)	0.062	(0.080)	0.439	(0.111)	0.212
Mother reads to child	-0.053	(0.016)	0.001	(0.015)	0.661
Learning materials	0.602	(0.066)	0.000	(0.071)	0.000
Mother stress (CES-D)	0.120	(0.081)	0.139	(0.072)	0.049
Mother works	0.102	(0.069)	0.138	(0.064)	0.097
Mother hours worked	4.823	(3.138)	0.124	(2.984)	0.097
Mother's earnings	10.652	(6.891)	0.122	(6.332)	0.070
Head's earnings	108.277	(7.736)	0.000	(3.336)	0.000

Notes: The column (2) specification of Tables 13–16 with standard errors clustered on project × parish (as in the main tables) and on the proposal, the level at which the score is assigned and treatment is determined (Abadie et al., 2023). With 34 clusters the analytic proposal-clustered standard errors are themselves subject to few-cluster bias, which motivates the wild cluster bootstrap in Table 10.

Table B3. Labor-market effects with community-level covariates

Outcome	Home visits baseline	Home visits +community X	Child care centers baseline	Child care centers +community X
Mother works	-0.257***	-0.268***	0.102	0.141***
	(0.061)	(0.055)	(0.069)	(0.044)
Mother hours worked	-8.918***	-17.035***	4.823	5.972***
	(1.817)	(1.972)	(3.138)	(2.098)
Mother stress (CES-D)	-0.689***	-0.889***	0.120	0.631***
	(0.116)	(0.108)	(0.081)	(0.163)
Mother's earnings	-37.278***	-58.387***	10.652	10.314**
	(8.897)	(5.567)	(6.891)	(5.032)

Notes: The column (2) specification of Tables 15 and 16 (baseline), and the same specification with community-level covariates added (parish population, consumption poverty, unsatisfied-basic-needs poverty, access to water and sanitation, and average schooling). Standard errors in parentheses, clustered on project × parish. Sample sizes are smaller in the community-covariate columns because the community variables are missing for some parishes. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.